

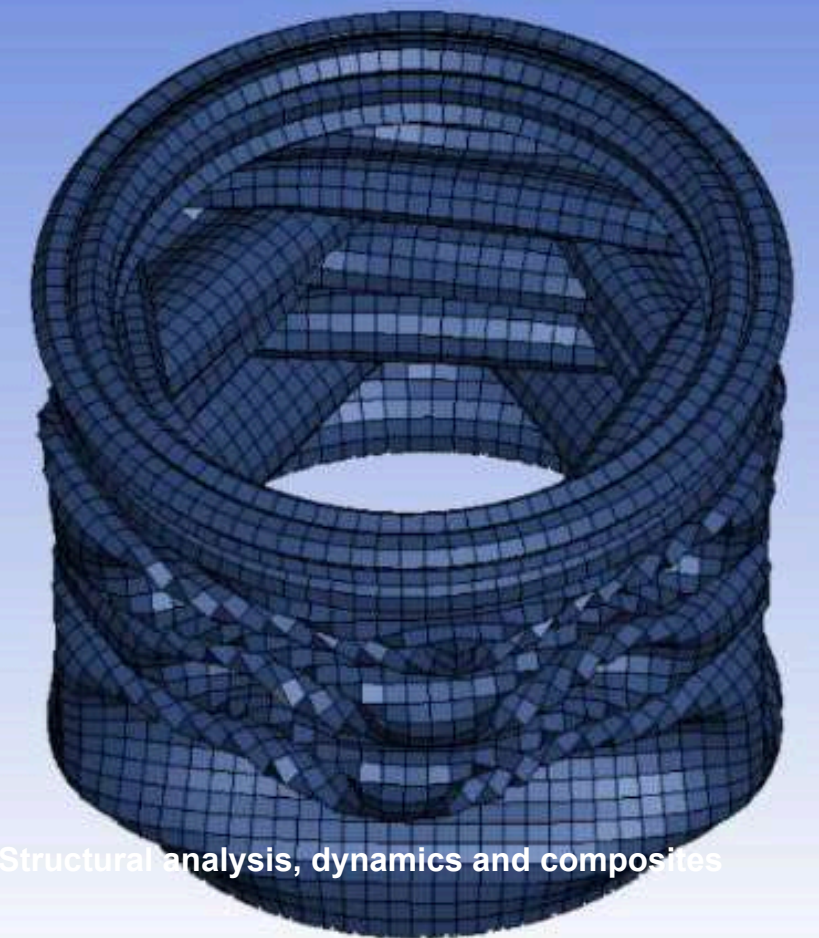
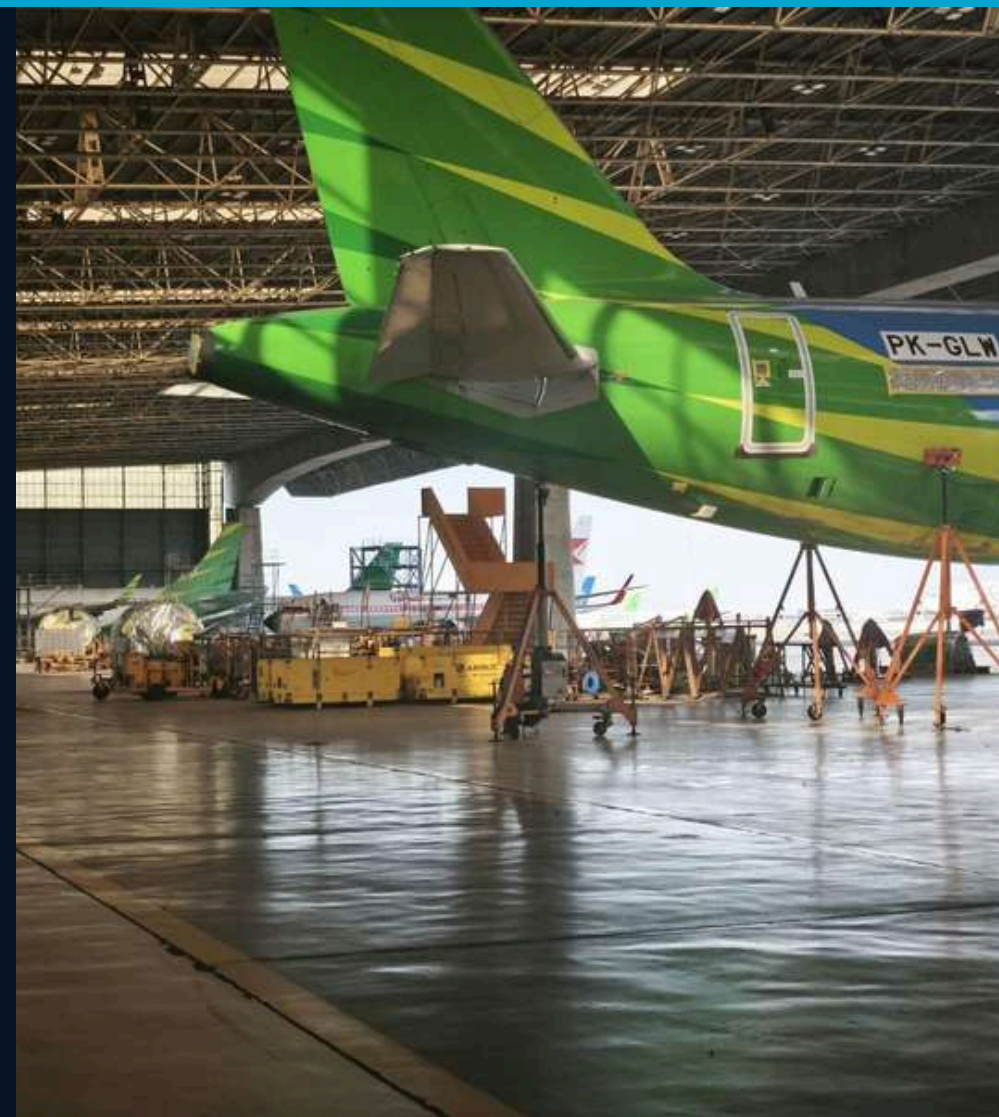
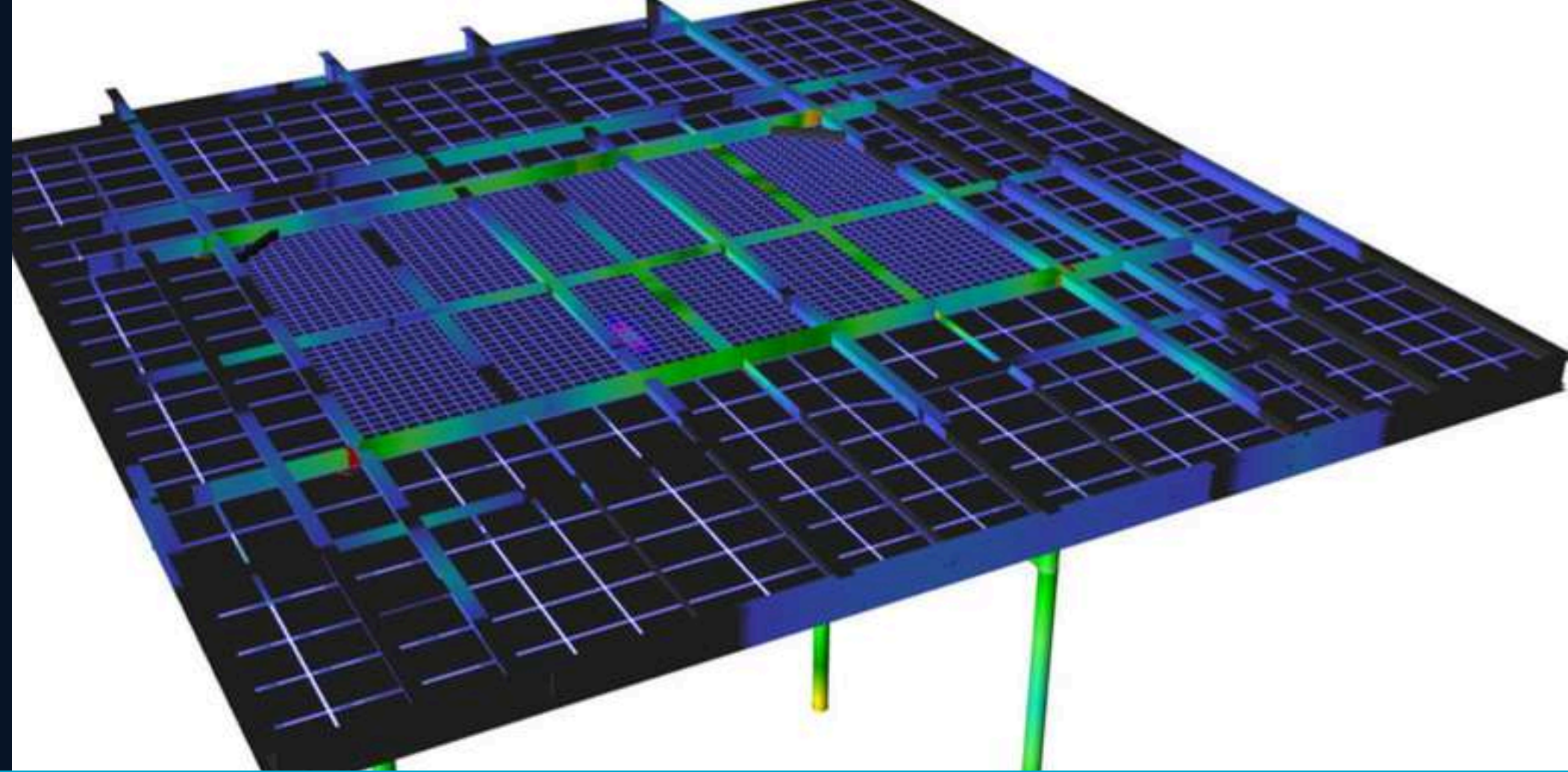
SELECTED WORK 2025-2026

ENGINEERING PORTFOLIO

HOTMAN JHONSON
ALEXANDER LIMBONG

Aerospace structures
Finite element analysis and automation

B.Sc. Aerospace Engineering
Institut Teknologi Bandung



Structural analysis, dynamics and composites



HOTMAN LIMBONG

Aerospace Engineering Graduate
Jakarta, Indonesia

FEA

STRUCTURES

PYTHON

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PROFILE

Structural analysis, verification,
and engineering automation

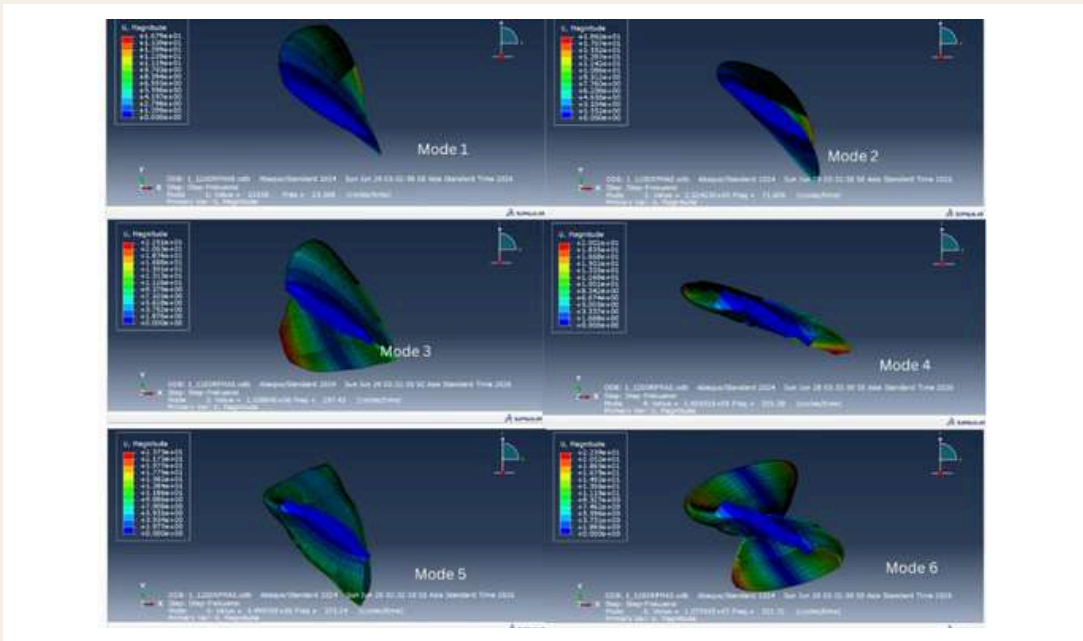
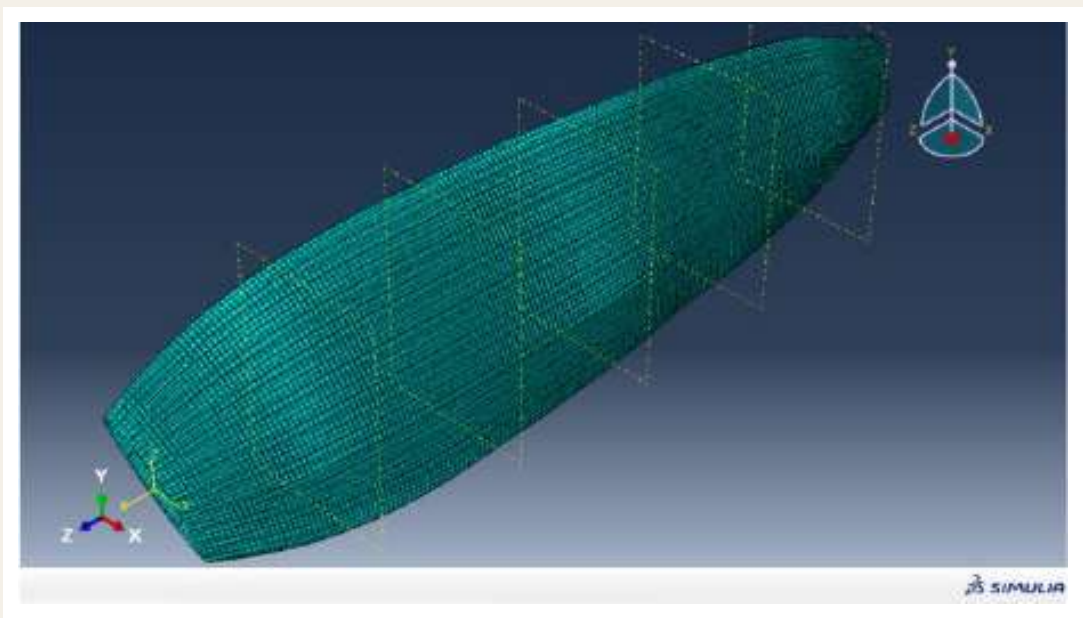
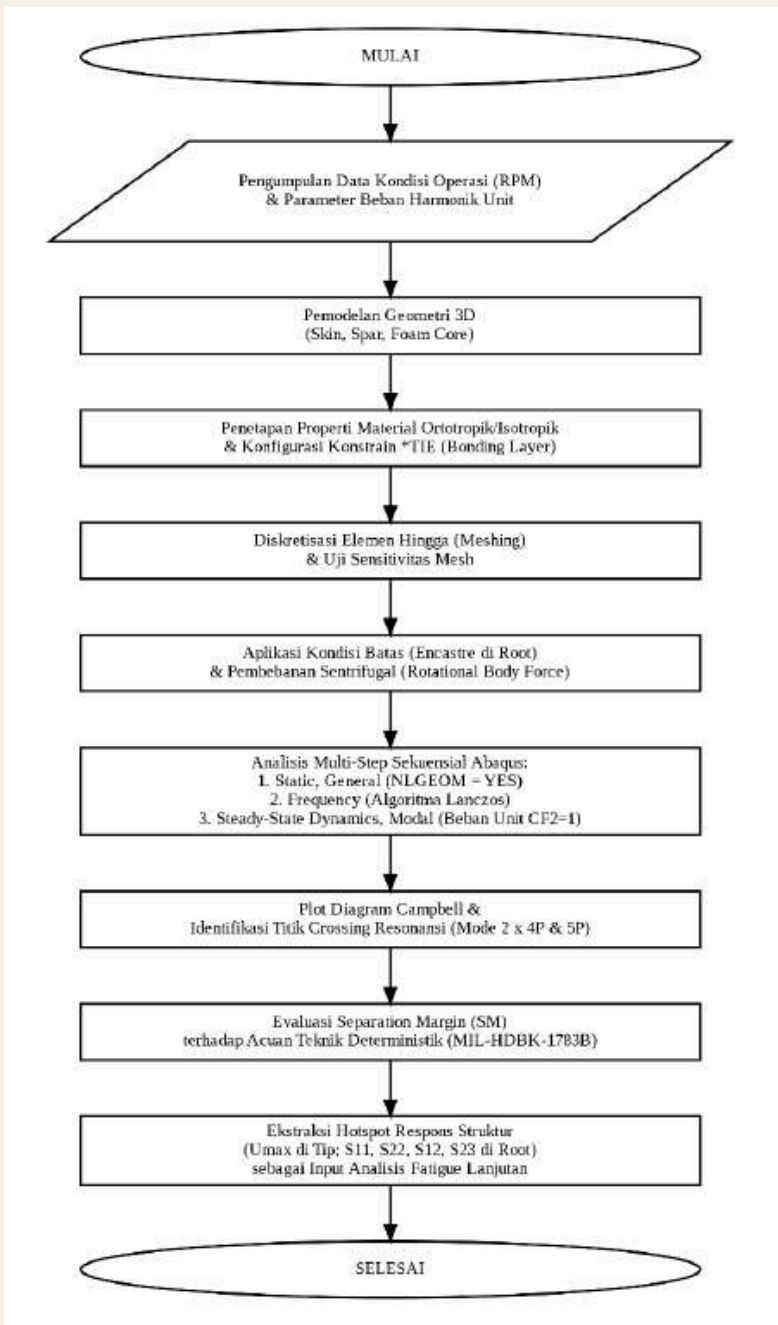
I am an Aerospace Engineering graduate with experience in aircraft and offshore structures. I have built and reviewed finite element models, traced load paths, assessed nonlinear and composite response, and automated simulation post-processing. I document assumptions and verification checks so another engineer can follow how I reached the result.

01	STRUCTURAL INTEGRITY	Global and local assessment, buckling, load paths and weld demand
02	NONLINEAR AND DYNAMICS	Material response, impact, structural dynamics and convergence
03	COMPOSITES	Laminate failure criteria, impact response and damage assessment
04	AUTOMATION	Python post-processing, data validation and engineering reports

ENGINEERING TOOLS	Abaqus, Bentley SACS, MSC Patran/Nastran, HyperMesh, Ansys CFX, LS-DYNA
DATA & REPORTING	Python, Excel, MATLAB and SQL
COURSE CODES	AE3103 Lightweight Structures AE4000 Vehicle Design AE4002 Airworthiness AE4012 High-Speed Aerodynamics AE4020 Finite Element Methods AE4021 Aircraft Loads AE4022 Impact Mechanics AE5026 Damage Tolerance

Dynamic response of a composite turboprop propeller blade

Rotational loading, engine-order excitation, and structural hotspot identification



RESEARCH QUESTION

How does centrifugal pre-stress change the blade modes, where do those modes approach the 1P-6P engine-order lines, and which structural regions carry the largest relative harmonic response?

FINITE ELEMENT MODEL

One Hamilton Standard 568F blade on an ATR 72-600 was represented as a sandwich assembly. Carbon/Epoxy AS4/3501-6 skin and spar use orthotropic composite properties, while Polyurethane FR-3708 represents the foam core. Skin layups taper from 14 to 4 plies along the span; TIE constraints join skin, spar, and core.

ANALYSIS SEQUENCE

Static General with NLGEOM develops centrifugal pre-stress. A Lanczos frequency step extracts six modes from 0 to 1,200 RPM. Campbell and separation-margin calculations prioritize operating points before Steady-State Dynamics, Modal evaluates the relative response with CF2 = 1.

0-1,200 RPM

analysed rotation range

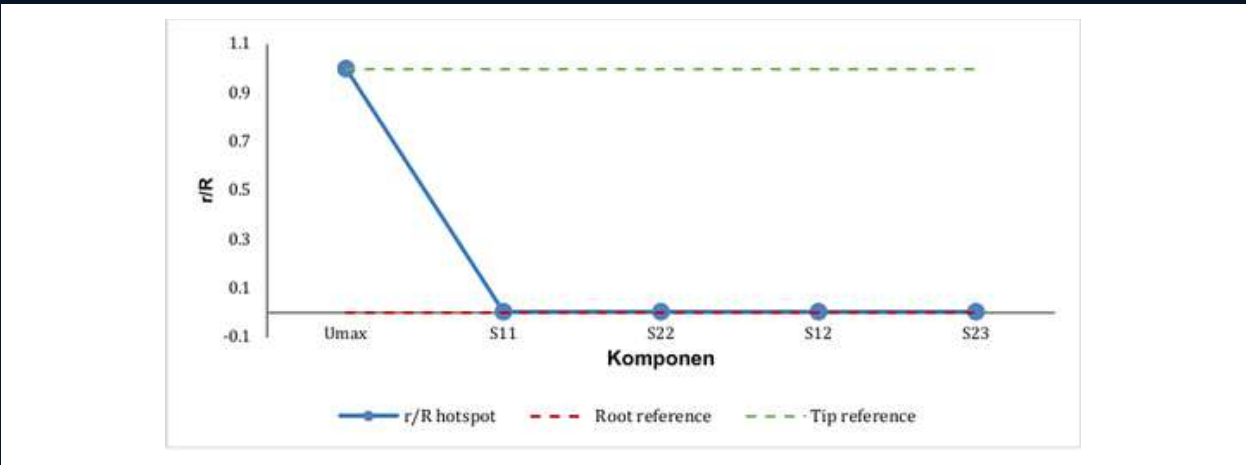
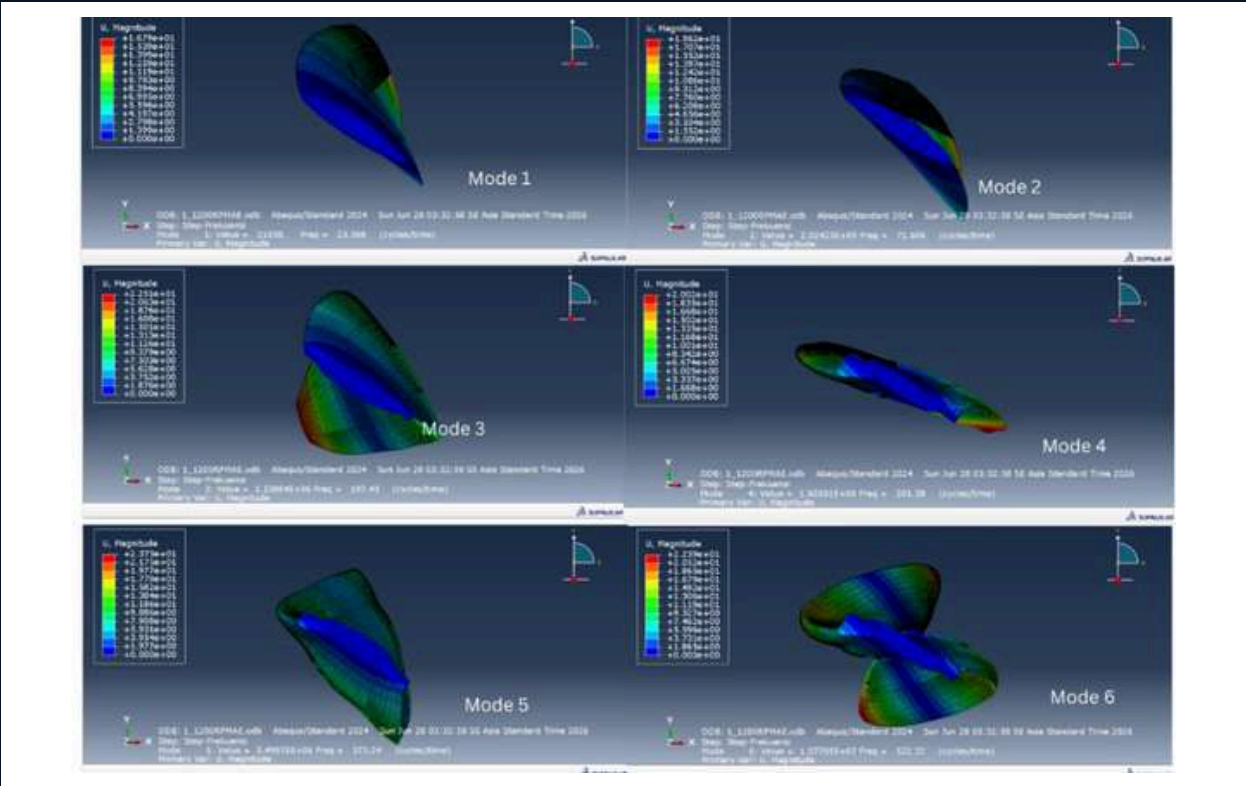
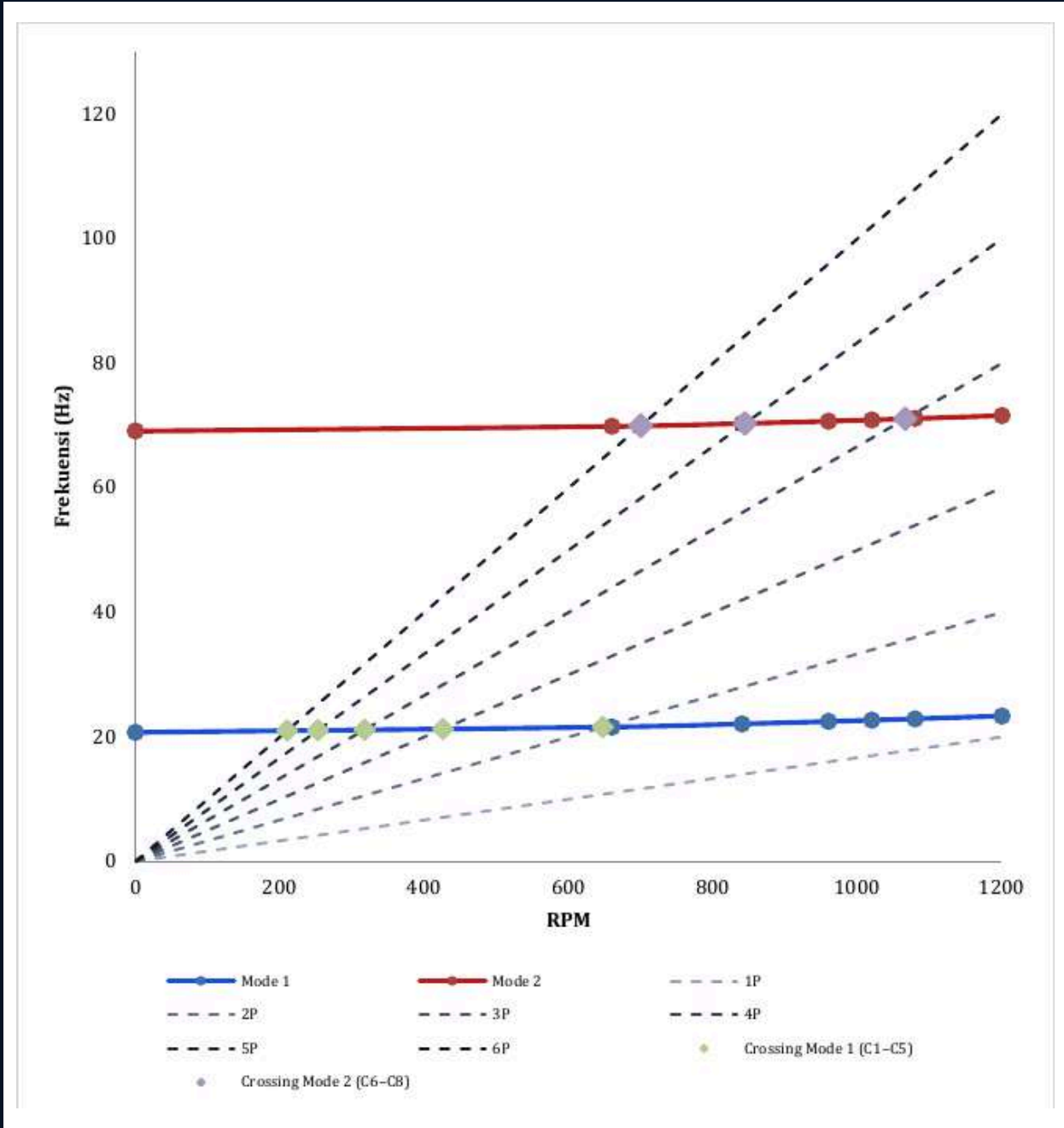
6 modes

tracked in the Campbell diagram

Validation: 21.892 Hz analytical vs 21.788 Hz Abaqus for a 3D cantilever, a 0.48% difference.

Mode 2 controls the closest engine-order interactions

The analysis separates frequency proximity, response amplification, and the physical location of structural hotspots.



MODAL TREND

Centrifugal stiffening raised Mode 1 from 20.740 to 23.368 Hz (+12.671%). Mode 2 increased by 3.597% and Mode 3 by 2.540%. Mode 2 remains in the 69-72 Hz band, where it approaches 5P and 4P excitation.

CRITICAL PROXIMITY

Mode 2 $\times$ 5P crosses at 844.263 RPM. Mode 2 $\times$ 4P crosses at 1,066.300 RPM. The smallest nominal separation margin is 0.489% at 840 RPM/5P, so that operating point receives the first harmonic-response check.

RESPONSE AND HOTSPOTS

With CF2 = 1, the SSD result represents response per unit harmonic load rather than an absolute flight load. A 1% damping sensitivity reduced the peak amplitude. Maximum displacement occurs at the tip; S11, S22, and S12 concentrate at the spar/root, while S23 concentrates in the foam core near the root.

The hotspot map identifies inspection priorities. It does not claim actual failure, fatigue life, or airworthiness without calibrated aerodynamic loading and experimental validation.

OFFSHORE HELIDECK

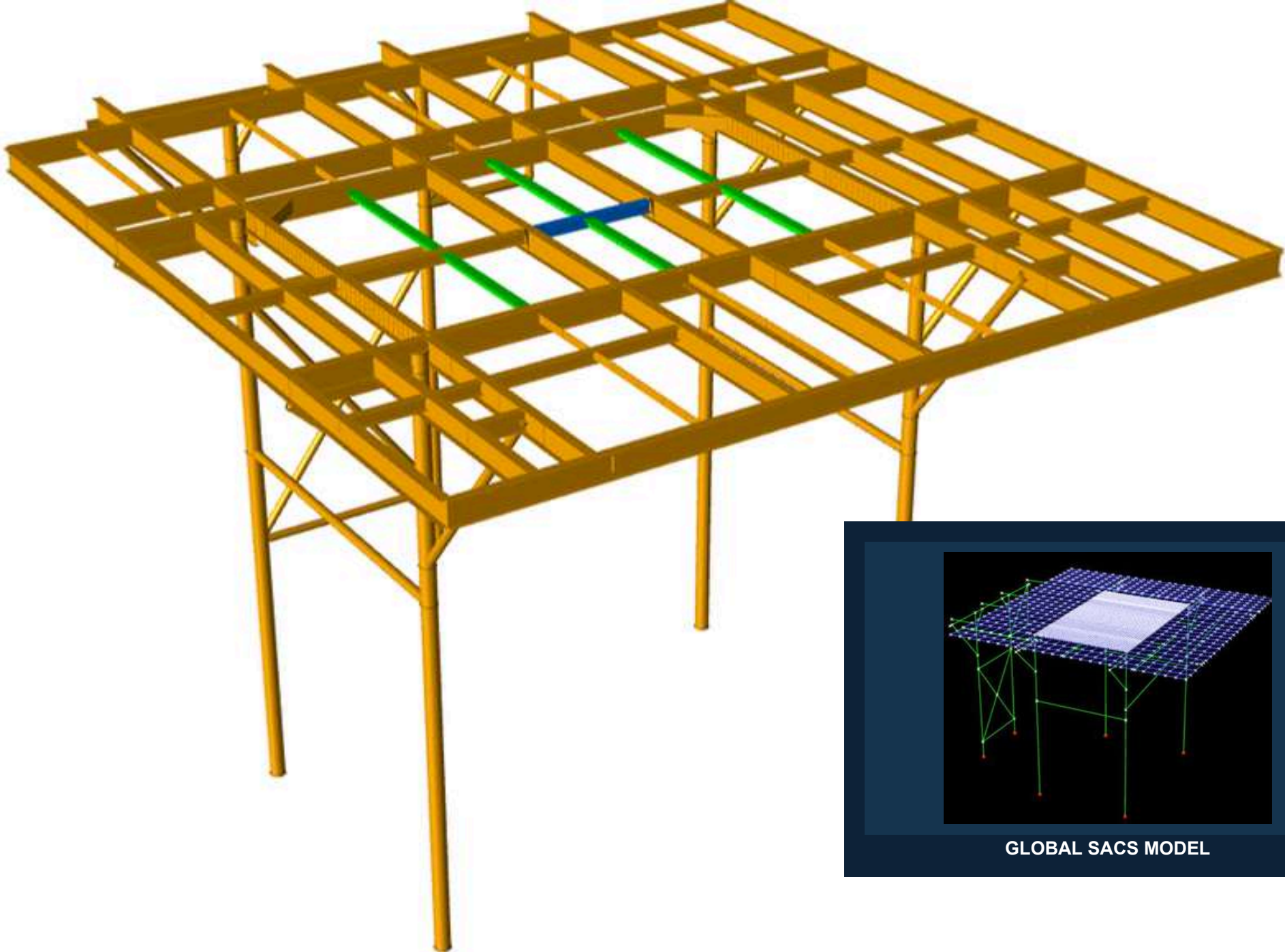
Global structural assessment

I supported the structural assessment of an offshore helideck for helicopter operations up to 5,000 kg MTOM. I reviewed member and tubular-joint unity checks, identified the governing load combinations, and traced the load path into the primary supports. The global model guided which connections needed detailed local analysis.

- BENTLEY SACS
- LOAD PATH
- UNITY CHECK

5000 kg

helicopter MTOM assessment basis



01

Member & tubular-joint unity checks

Screened utilization ratios and located the members that controlled the assessment.

02

Governing cases and support response

Compared reactions and force transfer for the critical operating combinations.

03

Member-upgrade and plate-thickening concepts

Compared member upgrades and plate thickening before detailed FE modelling.

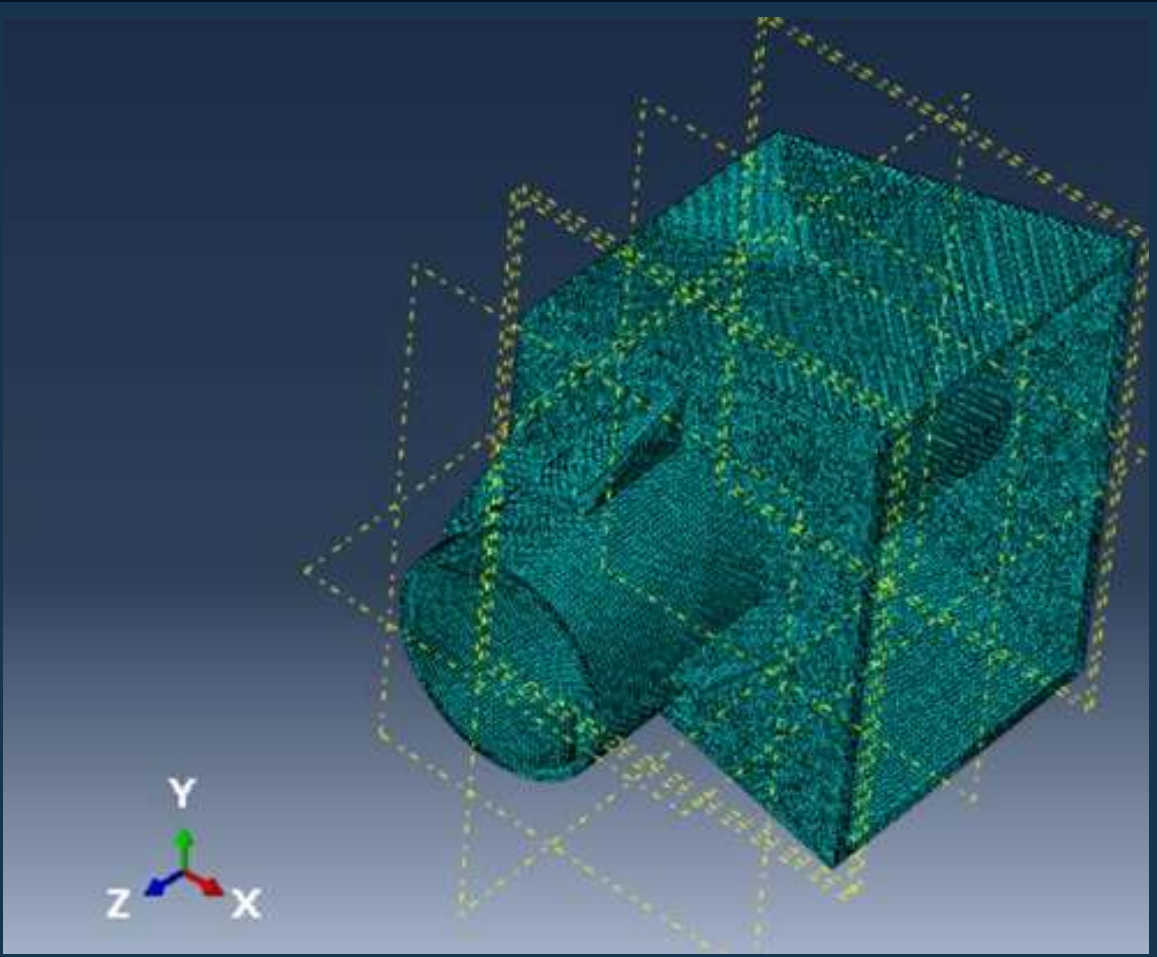
04

Deadweight change verified manually

The manual deadweight check differed from SACS by about 0.12%.

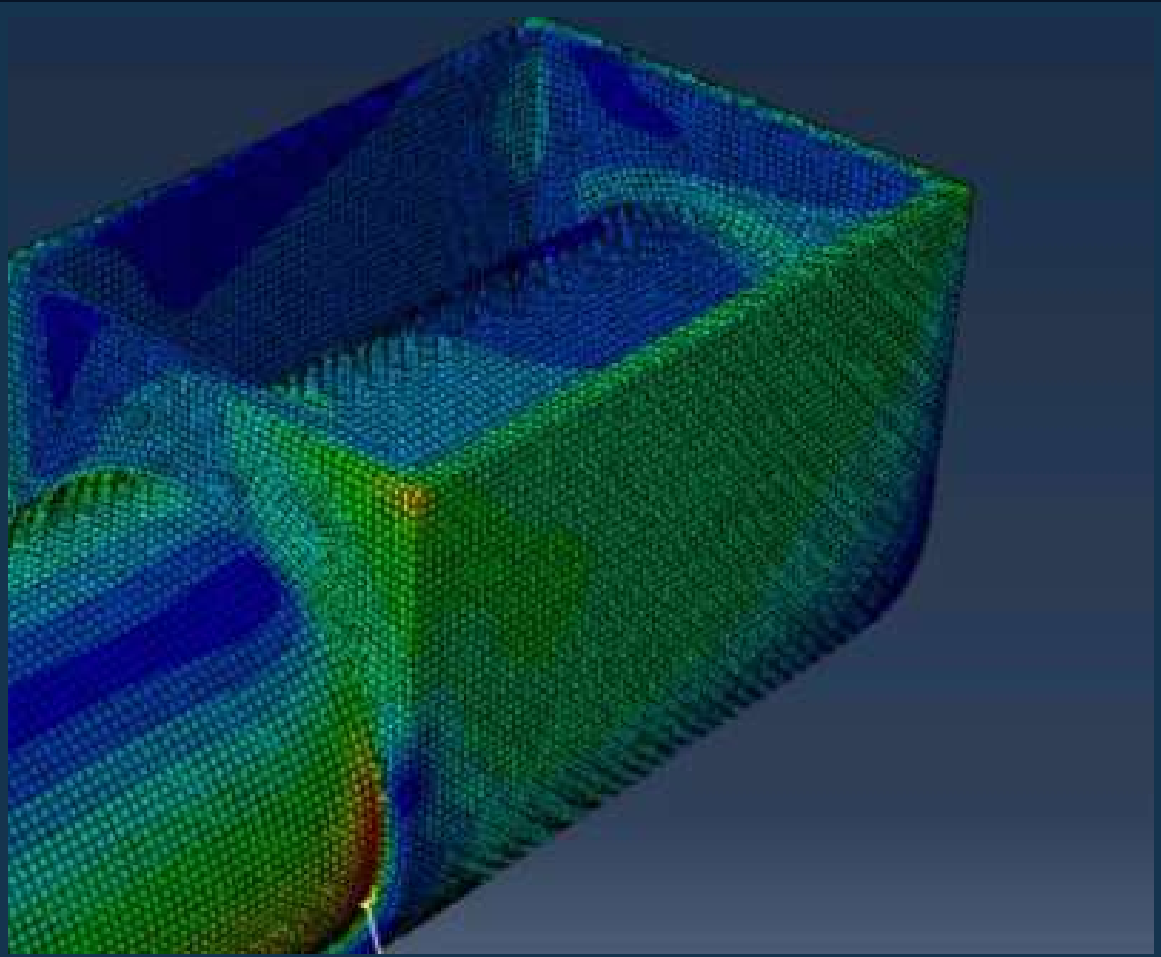
Local FE models and weld-demand extraction

C1 and C2 skid-support interfaces, nonlinear static assessment in Abaqus



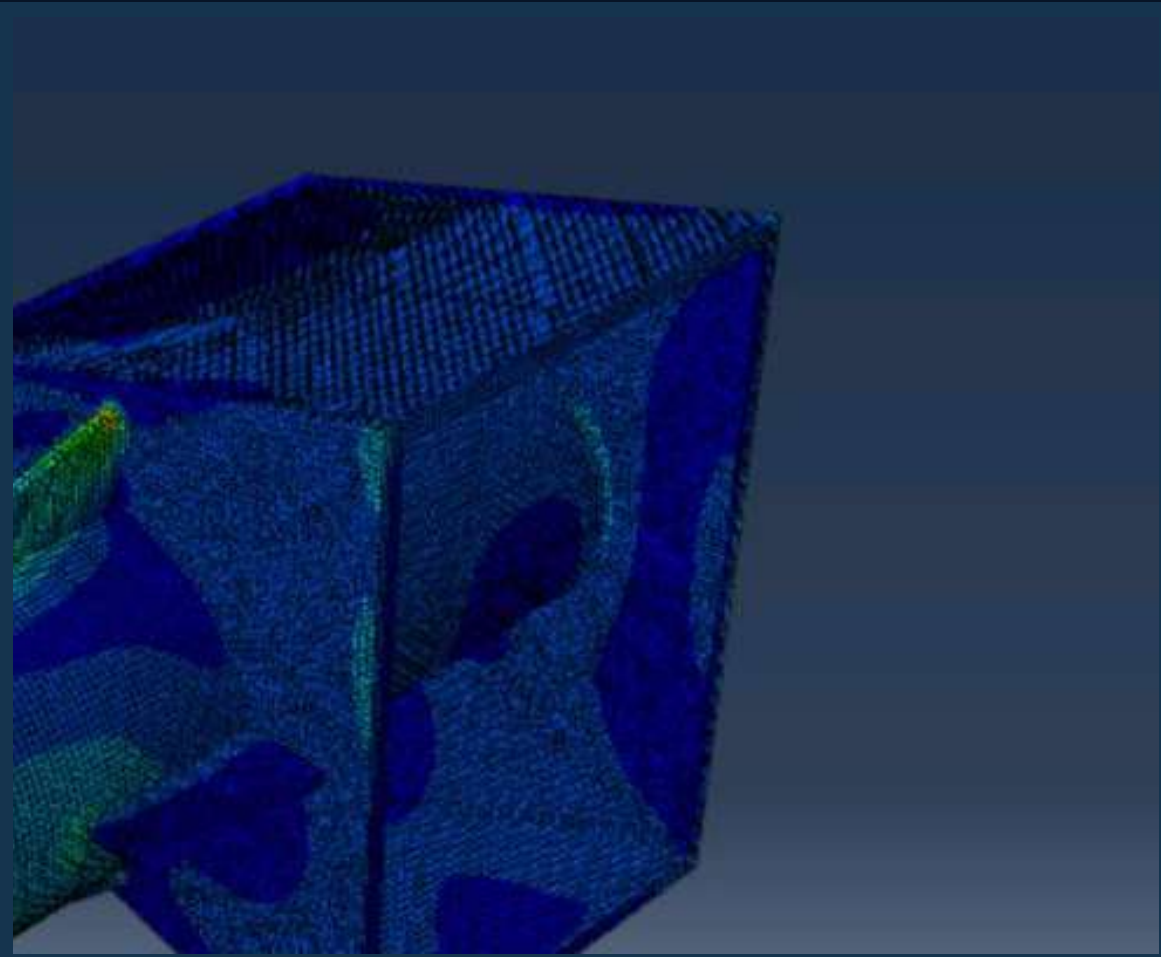
MODEL

I built the C1 and C2 assemblies, refined the mesh around support interfaces, and checked that further refinement no longer changed the engineering conclusion.



ASSESS

I ran the nonlinear static cases and reviewed stress, plastic strain, reactions, and local force transfer between connected parts.



EXTRACT

I extracted force and moment demand along each weld path, integrated the distributions, and checked endpoint closure before preliminary sizing.

10

C1 weld lines

22

C2 split-edge segments

≈0.12%

manual vs SACS deadweight difference

Abaqus

Four workstreams, one structural-analysis workflow

Jan-Apr 2026



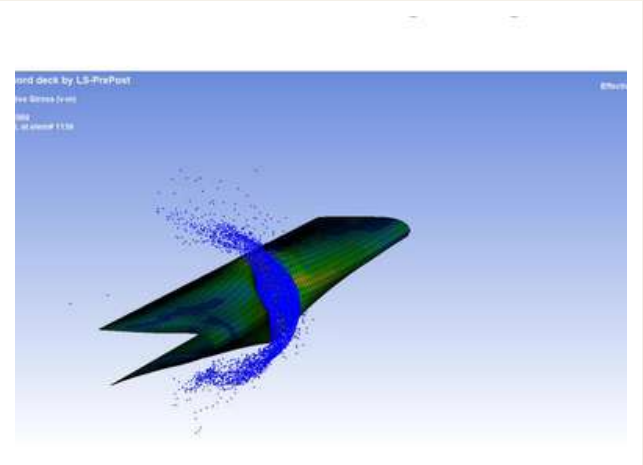
INTERNSHIP CONTEXT

I worked across finite element modelling, result interpretation, and Python-based post-processing. The assignments ranged from repeatable extraction of MSC Nastran results to explicit impact modelling and a geometry trade study. I documented inputs, checks, and outputs so each result could be reviewed by another engineer.



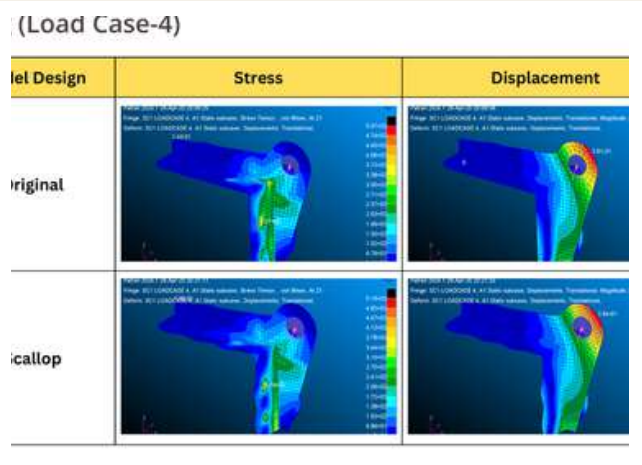
01 NEUBER AUTOMATION

Read .h5, .rpt, and .f06 output; correct local stress; report MoS in Excel.



02 COMPOSITE FAILURE

Evaluate every element and ply against seven laminate failure criteria.



03 BIRD STRIKE

Compare Al7075-T6 and CFRP leading edges under an 85 m/s SPH impact.

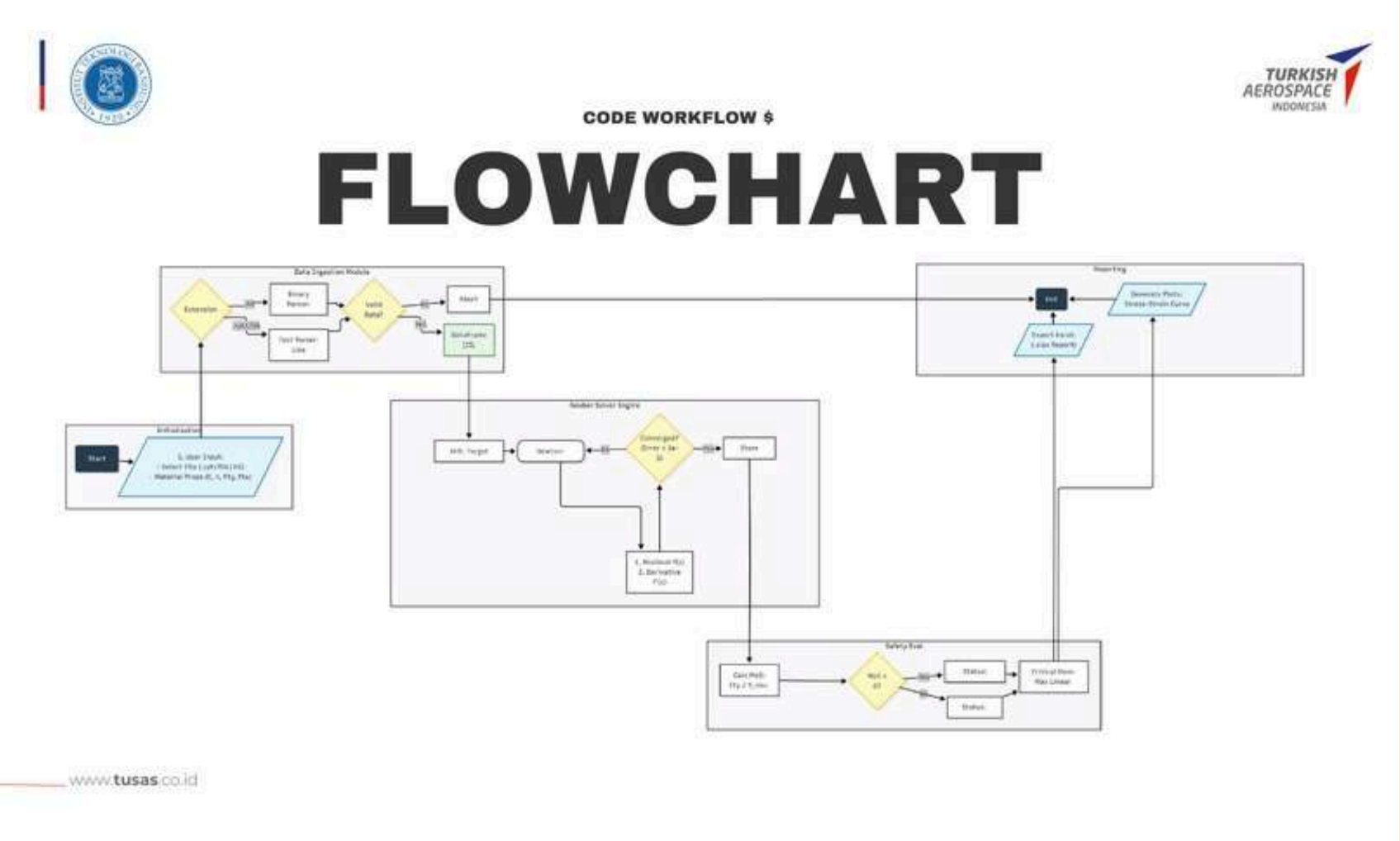
04 SCALLOPED FITTINGS

Quantify the mass saving and structural penalty of front and rear lug cut-outs.

TOOLS Python · PyTables · Pandas · NumPy · SciPy · Excel · Patran/Nastran · HyperMesh · LS-DYNA

From Nastran .h5 output to a reviewable engineering report

The script keeps the simulation result, nonlinear correction, and final decision traceable to the same element.



Automated data path from solver files to the engineering workbook

01 EXTRACT

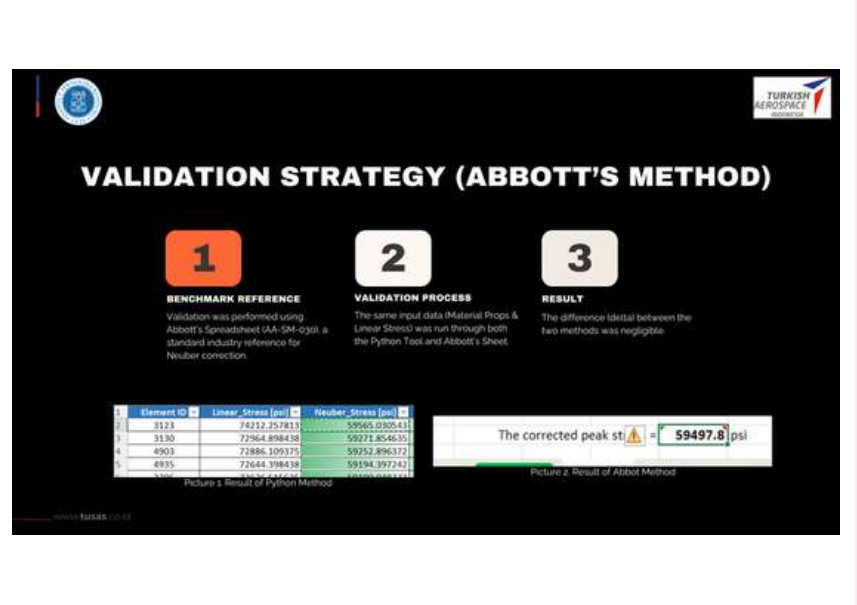
glob locates .h5, .rpt, and .f06 files. PyTables reads the MSC Nastran HDF5 hierarchy, including element IDs and linear stresses. Pandas and NumPy organize the fields into calculation arrays.

02 CORRECT AND VERIFY

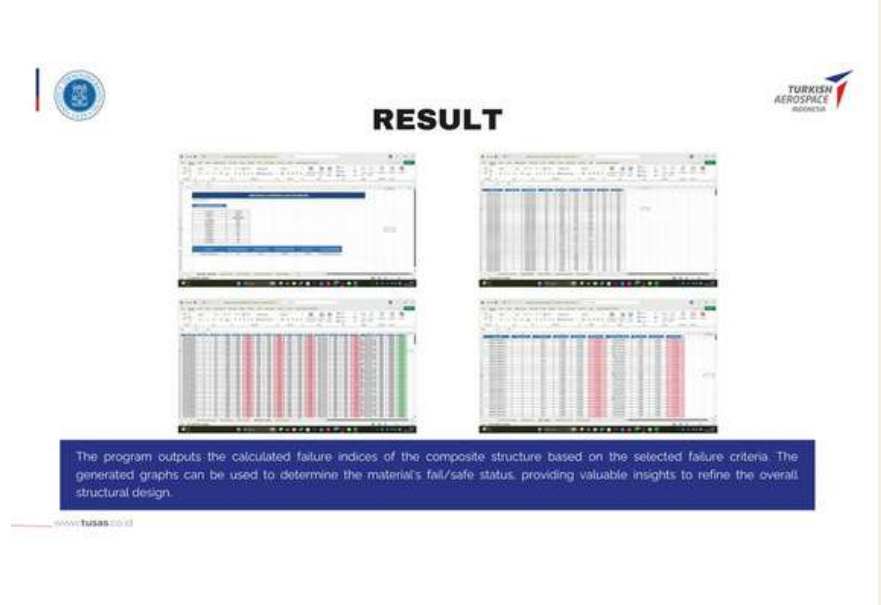
SciPy solves the Neuber and Ramberg-Osgood equations until the nonlinear residual is below 1×10^{-5} . The corrected stress drives $MoS = Fty / Sneu$. The Python result matched Abbott AA-SM-030 with a negligible reported difference.

03 REPORT AND EXTEND

The tool highlights the critical element and writes formatted Excel tables and stress-strain plots. A related evaluator preserves load case, element, ply, and criterion while checking seven composite failure criteria.



Abbott AA-SM-030 benchmark



Ply-by-ply failure output

RESIDUAL < 1E-5

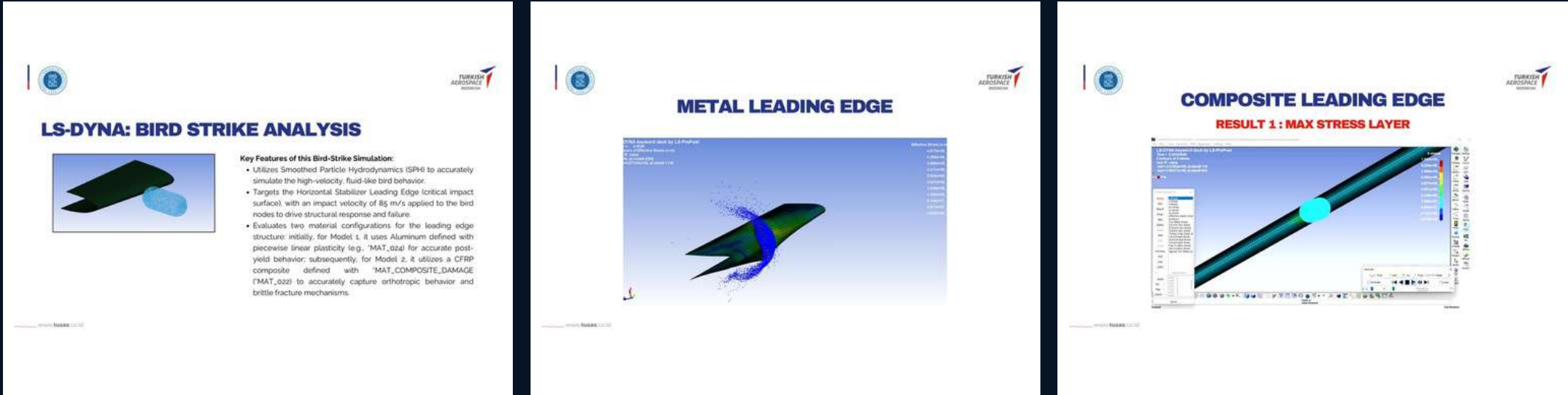
CRITICAL ELEMENT

7 FAILURE CRITERIA

EXCEL OUTPUT

Bird strike and scalloped-lug trade studies

Two different questions: impact survivability of a leading edge and weight reduction of titanium fittings



Boeing 737NG maintenance-defect analysis

Operational records were converted into comparable defect patterns by fleet age and aircraft area.



Internship presentation and discussion

METHOD

I cleaned inconsistent labels, mapped each record to an aircraft zone and fleet-age group, and calculated counts and percentages in Excel. The same taxonomy was applied across the three age groups so the comparison remained consistent.

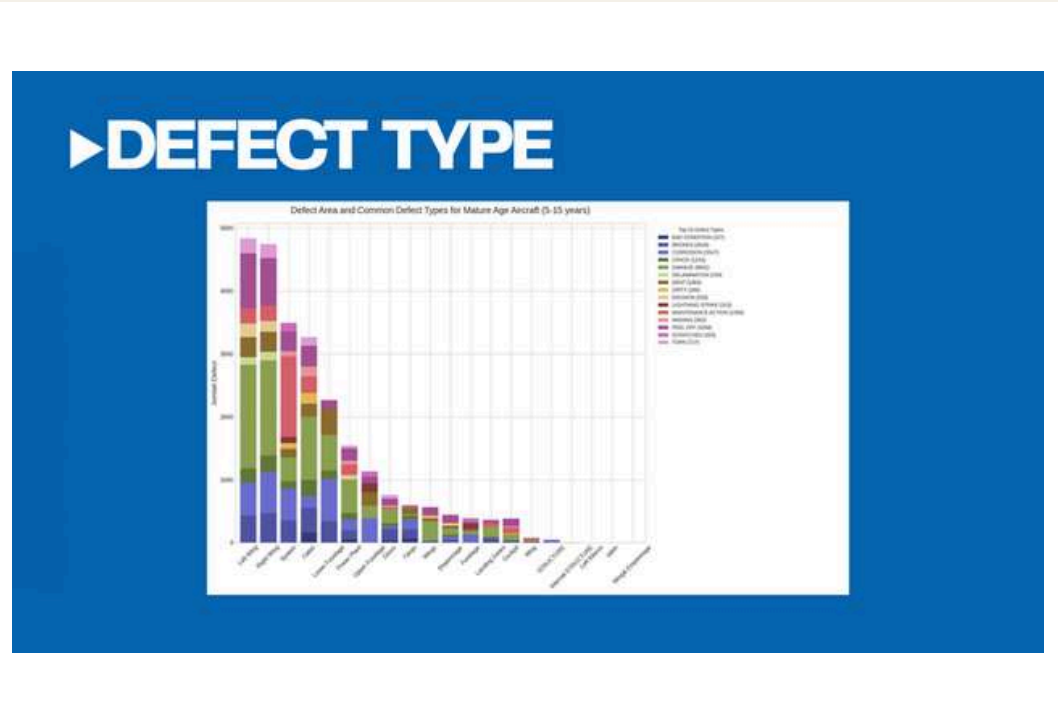
40,000+
defect records reviewed

3
fleet-age groups

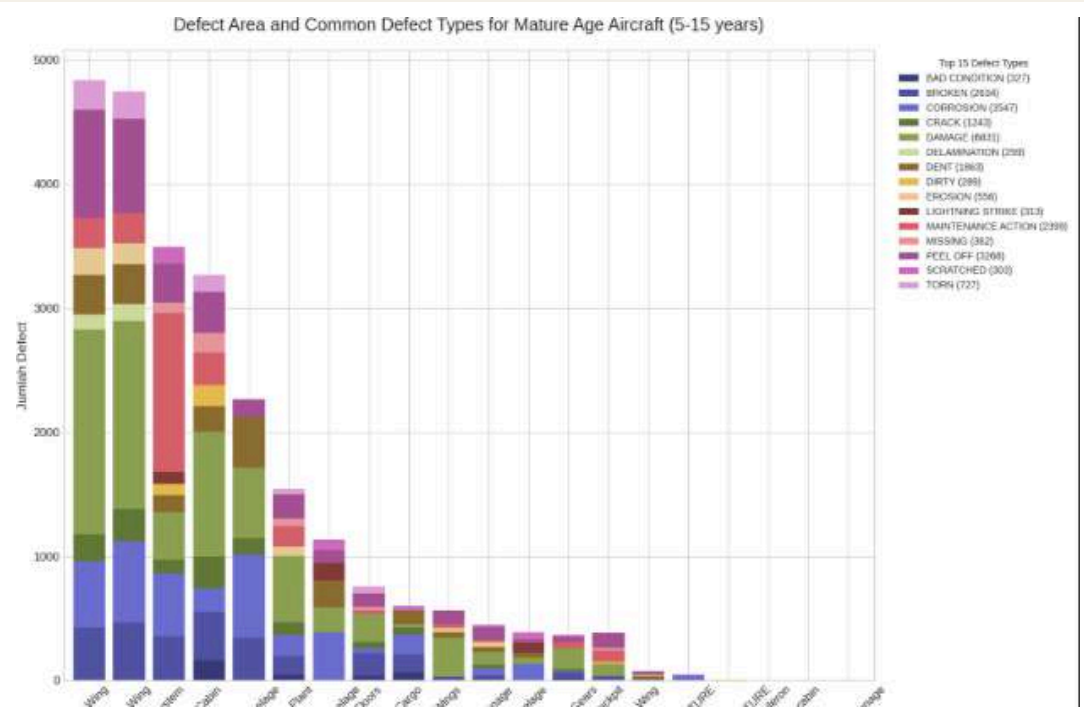
ENGINEERING INTERPRETATION

The analysis compared recurring defect categories and their physical locations. Instead of treating every record as an isolated event, the final presentation showed where maintenance attention clustered as the fleet matured.

737NG
aircraft family



Young fleet: defect type



Mature fleet: defect area

OUTPUT

I prepared the charts and final review material for maintenance stakeholders. The work also gave me practical exposure to MPD/MP analysis, MRO workflows, operational risk, HSE, and aviation security.

Core tools: Excel data cleaning, category mapping, pivot analysis, charts, and presentation of maintenance findings

Operational reporting across regional work units

Jul-Aug 2025



Team documentation

	C	D	E	F	
	Agustus				
	Jenis Simpanan	Nama Nasabah	Nomor Rekening	Tanggal Masuk Komitmen	Noni tan
	Giro			18/08/2025	
	Giro			27/08/2025	
	Giro			26/08/2025	
	Giro			29/08/2025	
	Giro			29/08/2025	
	Giro			28/08/2025	
	Giro			29/08/2025	
	Giro			21/08/2025	
	Giro			15/08/2025	
	Giro			19/08/2025	
	Giro			29/08/2025	
	Giro			28/08/2025	
	Giro			26/08/2025	
	Giro			28/08/2025	
	Giro			26/08/2025	
	Giro			27/08/2025	
	Giro			29/08/2025	
	Giro			27/08/2025	
	Tabungan			27/08/2025	
	Deposito			06/08/2025	

Names, account numbers, and values are covered

[illegible]

Merchant names and IDs are covered

REPORTING ROUTINE

I consolidated daily, weekly, and monthly Excel reports from regional work locations. Before reporting, I checked missing fields, inconsistent naming, duplicated entries, and whether the requested reporting period matched the source.

ANALYTICAL SUPPORT

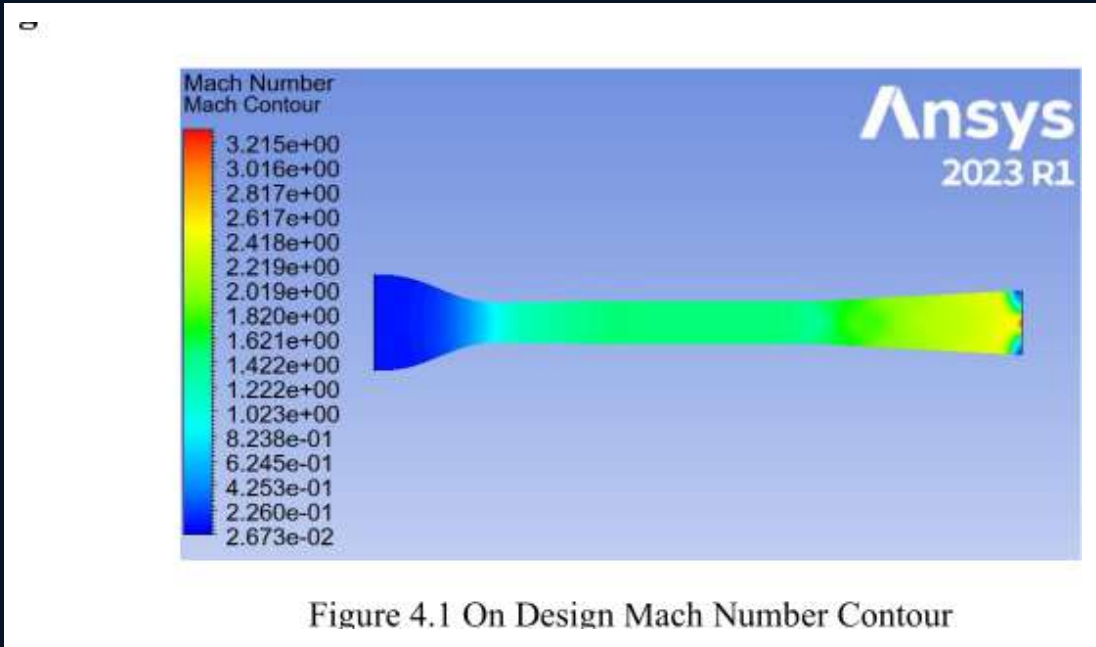
I supported performance monitoring, analytical-model testing, data-dictionary updates, and process reviews for the Mass Retail Fund team. The work required careful reconciliation because a small reporting error could alter a management summary.

DATA HANDLING

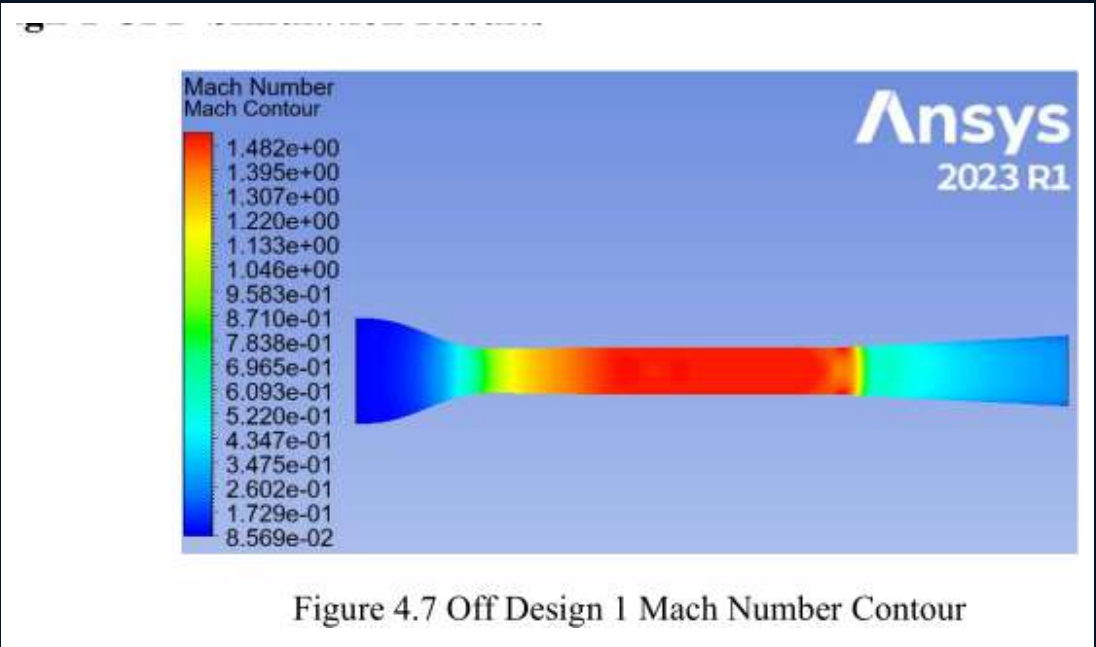
Excel remained the main working tool for validation, lookup, filtering, reconciliation, pivot summaries, and management-ready tables. Only identity fields and financial values are masked in this portfolio; the report structure remains visible.

Supersonic wind-tunnel design at Mach 1.5

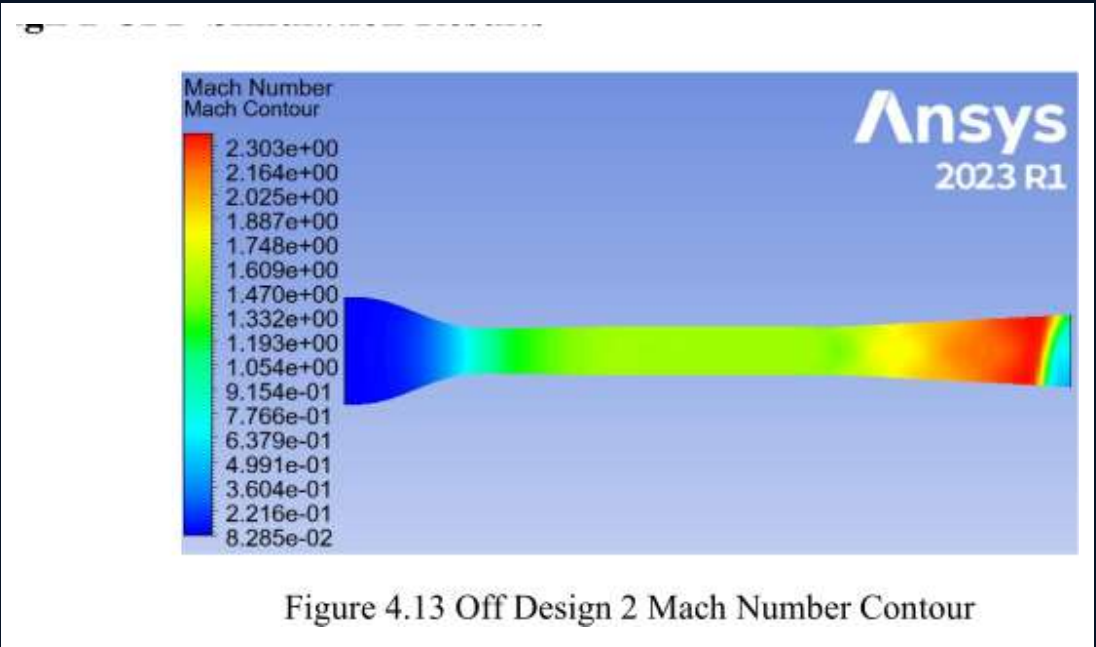
Quasi-one-dimensional analysis, Method of Characteristics, structured meshing, and ANSYS CFX comparison



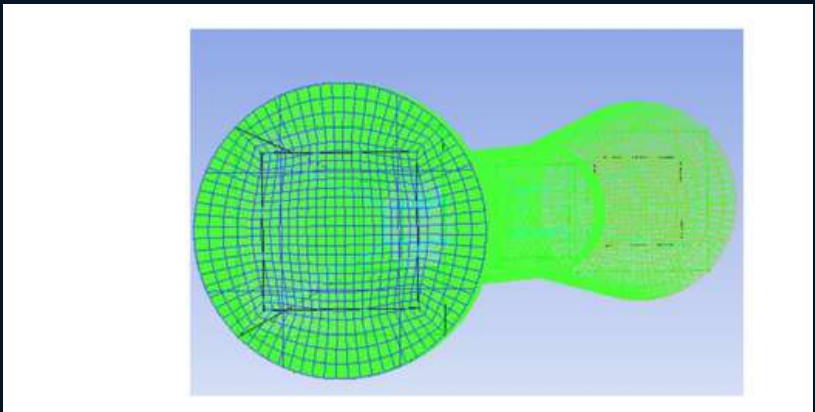
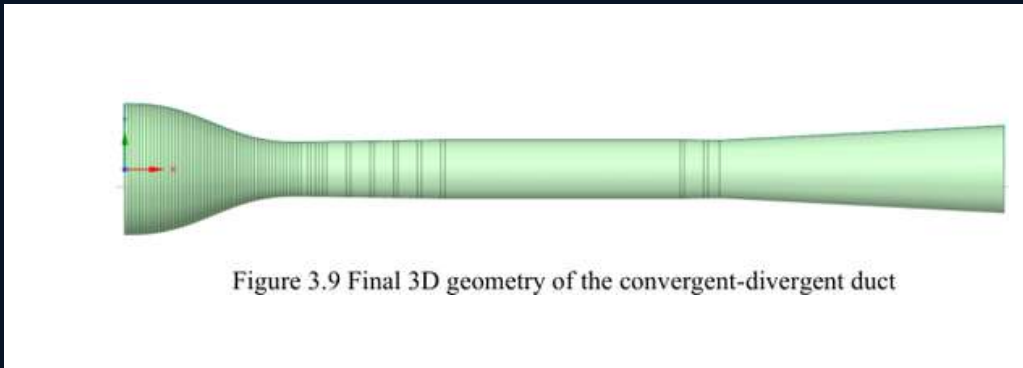
ON DESIGN Test section reaches the assigned Mach 1.5 condition.



OFF DESIGN 1 The normal shock moves toward the end of the test section.



OFF DESIGN 2 Higher back pressure pushes the shock upstream from the ideal exit.



DESIGN METHOD

The study sized the inlet, first throat, test section, second throat, and diffuser from compressible-flow relations. A fifth-order polynomial defined the convergent section and the Method of Characteristics defined the minimum-length supersonic nozzle. SpaceClaim generated the volume, ICEM CFD created the O-grid, and CFX solved the on-design and two off-design cases.

Mach 1.5

target test-section flow

15 kg/s

design mass flow

1.546 m

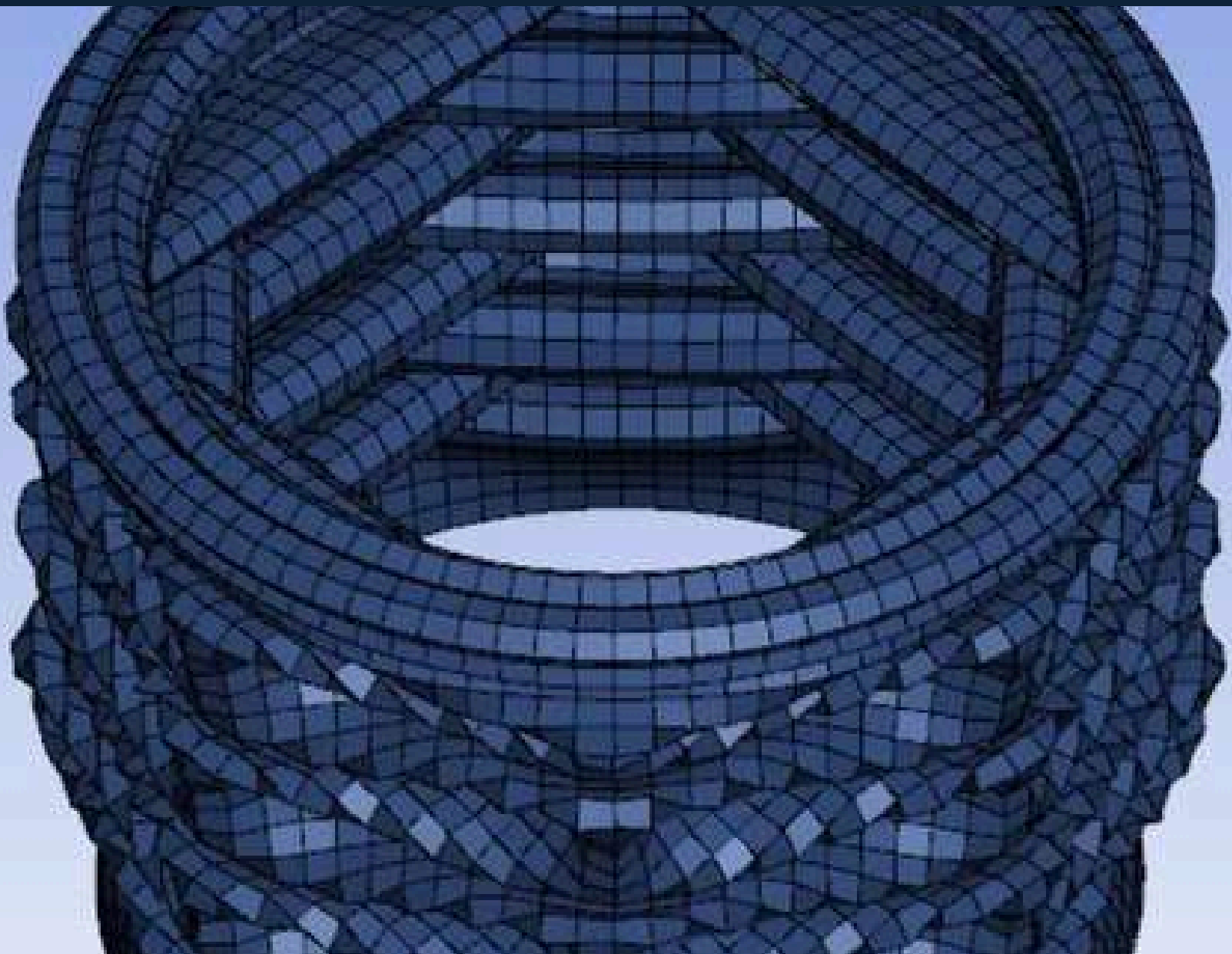
reported tunnel length

OFF-DESIGN RATIOS

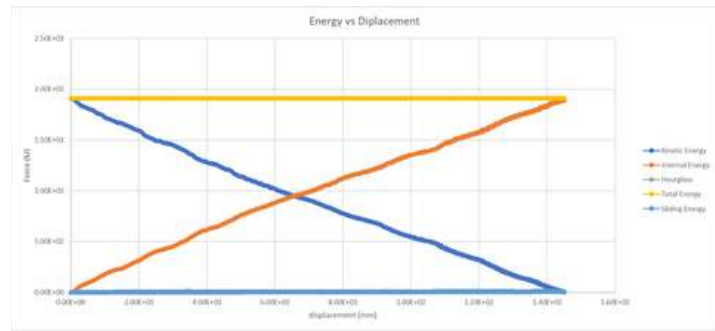
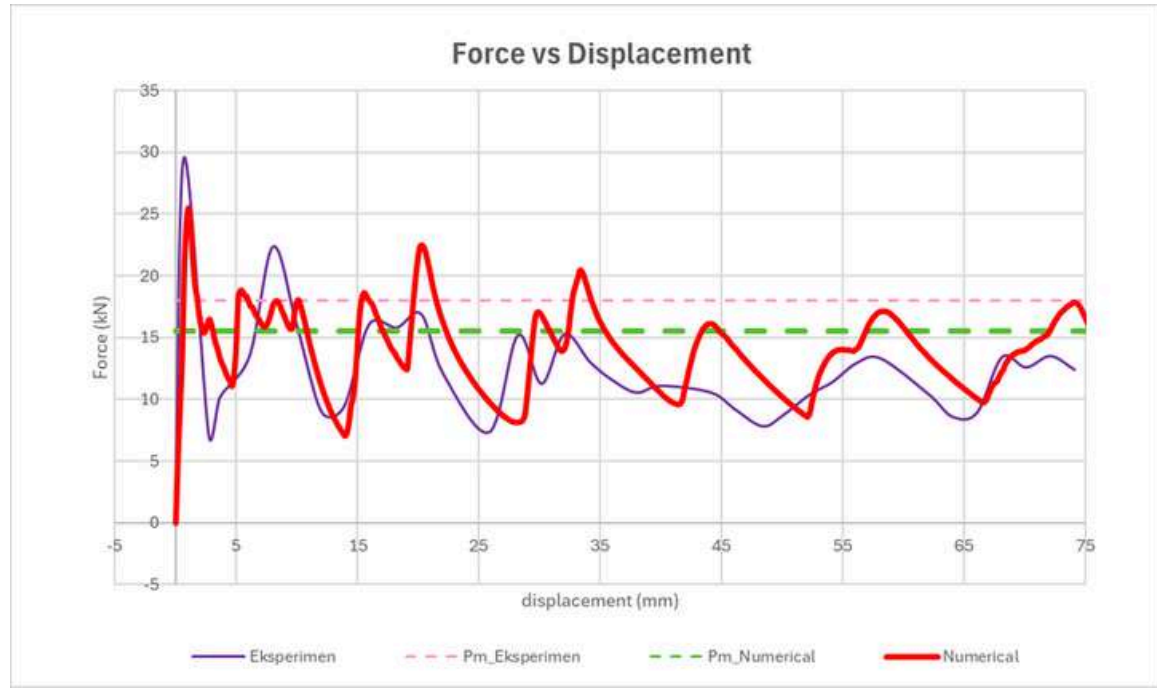
$$A_2^*/A_1^* = 1.03 \text{ and } 1.08$$

The comparison checks whether the analytical shock location and pressure recovery remain consistent with the CFD response.

CRUSHING RESPONSE OF A THIN-WALLED STRUCTURE



Our team modelled a 200 mm Al6063-T6 circular crash box in LS-DYNA and loaded it axially with a rigid flat-wall impactor. The 2 mm mesh reproduced progressive folding. We checked force-displacement response and energy balance, including low hourglass and sliding energy, before comparing the model with theory and experiment.



2 mm

selected mesh size

13.03 / 29.54

numerical mean / peak force (kN)

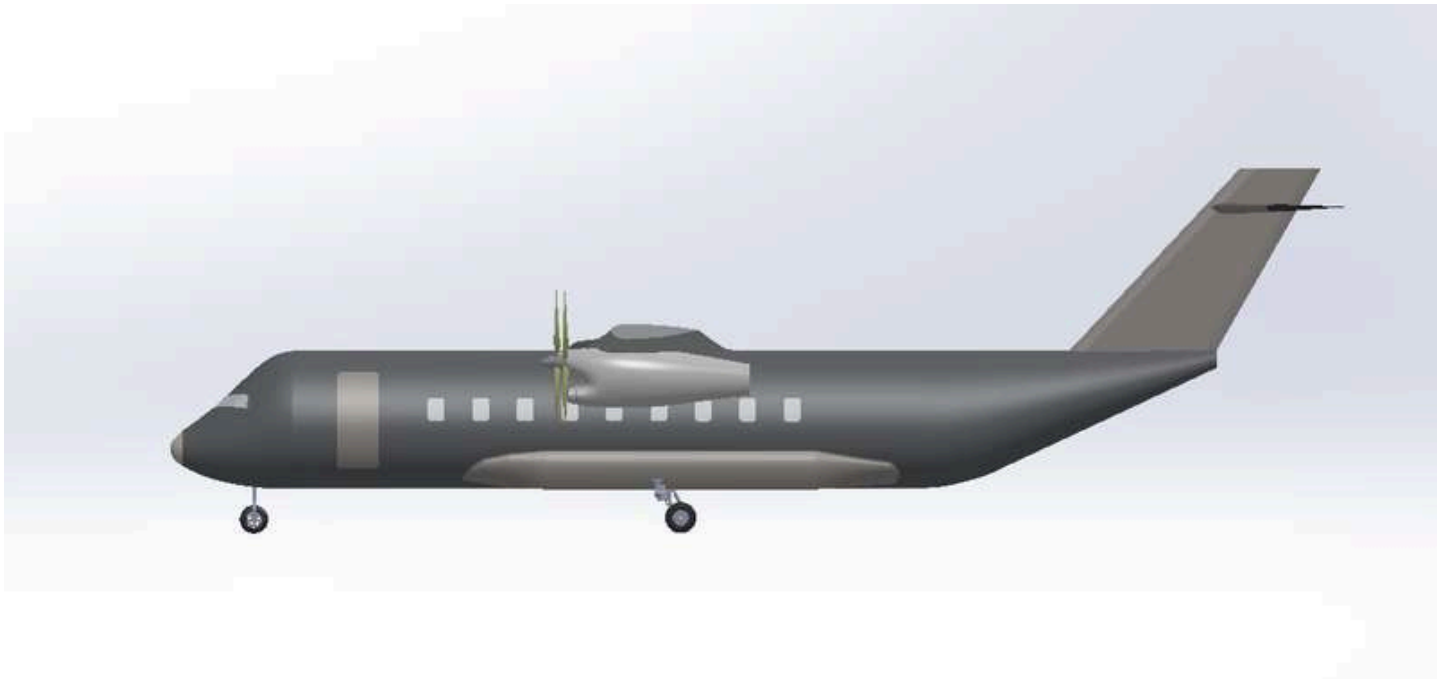
15.63 kN

analytical mean crushing force

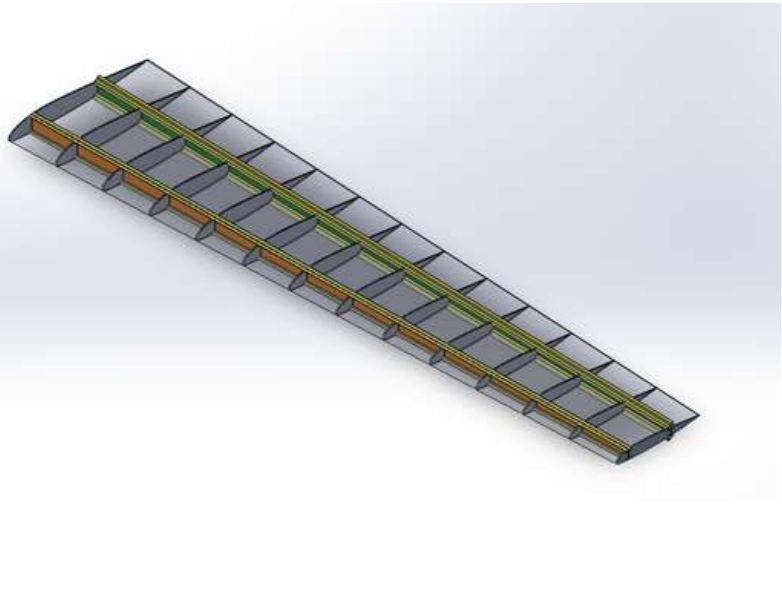
17.98 kN

experimental mean crushing force

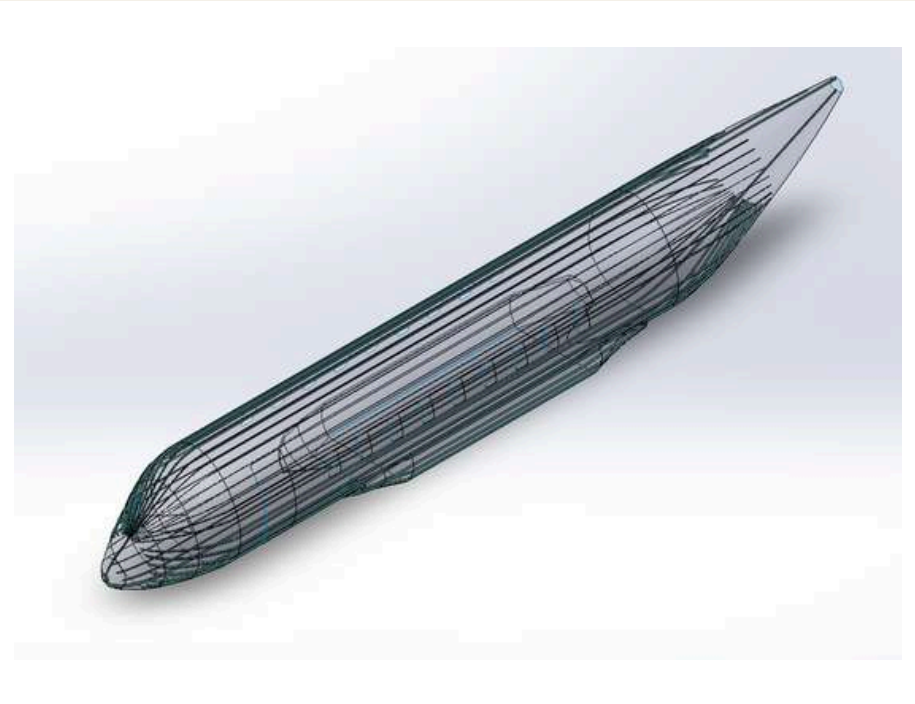
LAHAR-001 hybrid-electric regional commuter



Overall aircraft concept



Wing structural layout



Fuselage structural layout

CONCEPT AND PERFORMANCE

Our team translated the mission into a 19-passenger, high-wing aircraft with a T-tail, retractable landing gear, and a hybrid-electric propulsion concept. The final sizing reported a 928.76 km range, 301.87 km/h cruise speed, and 486.67 m takeoff distance.

STRUCTURAL SIZING

The primary structure was checked at the ultimate load factor $n = 5.7$ using CFRP T800S and Al-Li 2198. The report records positive margins for the main components, with 603,912 Nm wing-root bending moment and 172,688 N shear per side.

19

passengers

928.76 km

reported range

301.87 km/h

reported cruise speed

486.67 m

reported takeoff distance

8,508 kg

final reported MTOW

$n = 5.7$

ultimate structural case

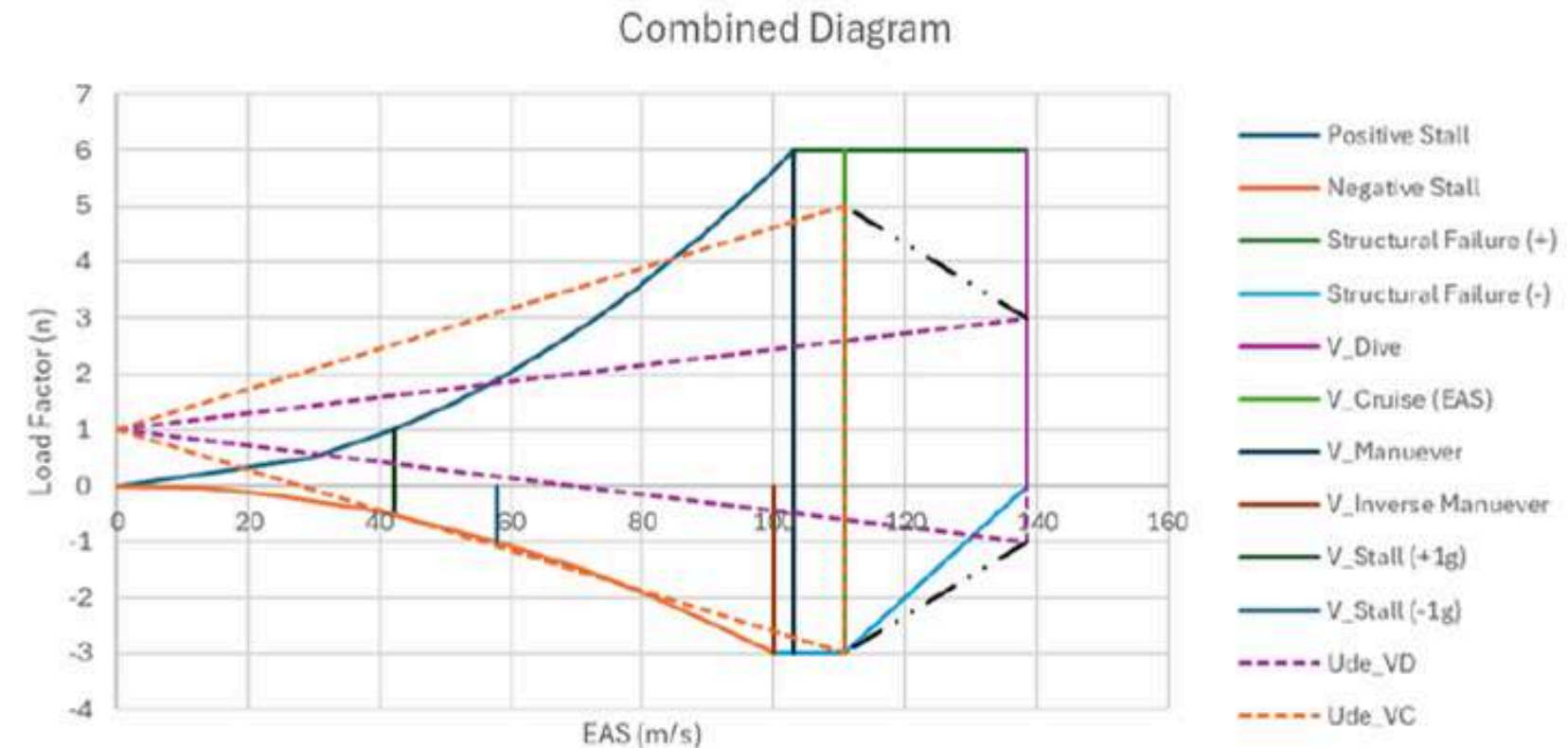
B-NDAR LOAD ANALYSIS

Aiska Vairana and I built maneuver and gust envelopes for the B-NDAR counter-insurgency aircraft using MIL-A-8861B. We distributed aerodynamic and inertia loads along the wing and fuselage, then integrated them into shear-force and bending-moment diagrams.

MIL-A-8861B

+6 / -3 G

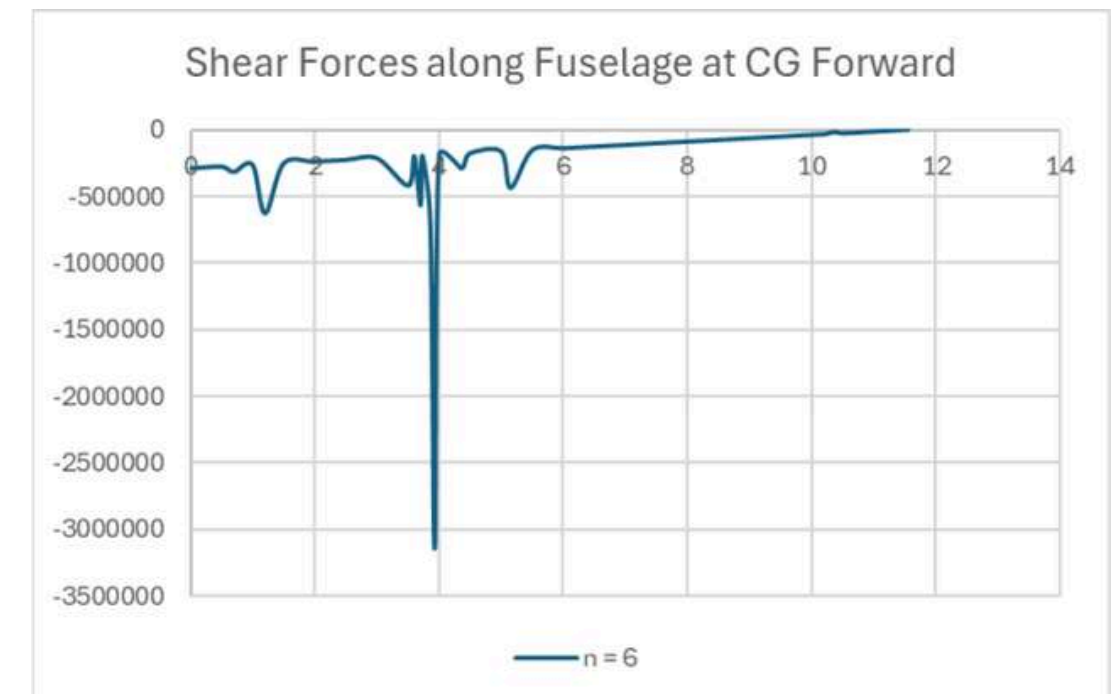
- +344,682 N and -172,341 N total lift for the +6 g and -3 g limit cases
- A CG shift of ± 0.25 m changed the reported root bending moment by about $\pm 86,171$ Nm
- The wing root and wing-fuselage junction governed the internal-load review



Combined maneuver and gust envelope



Wing internal shear



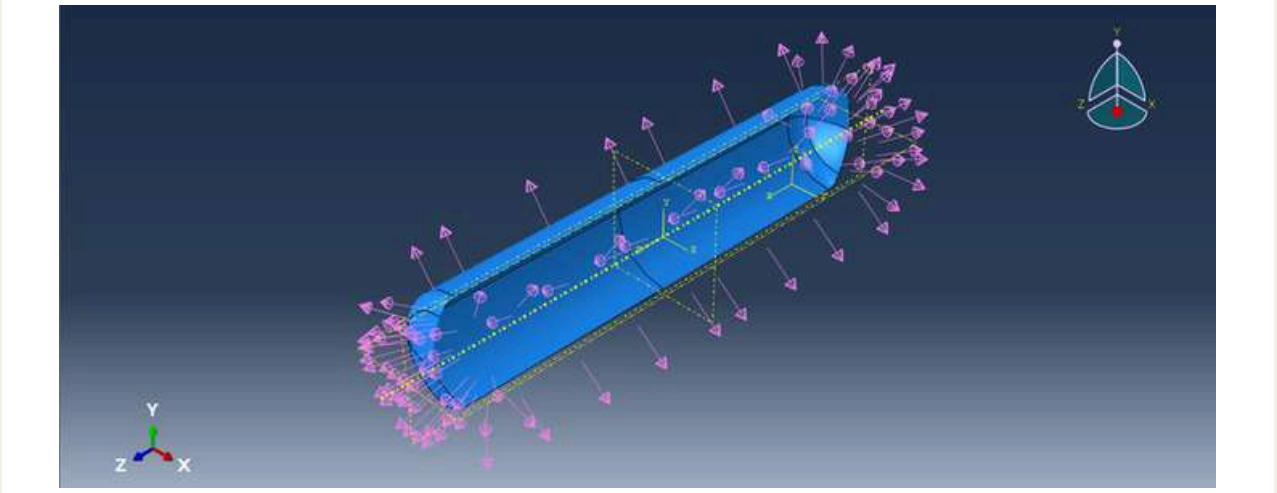
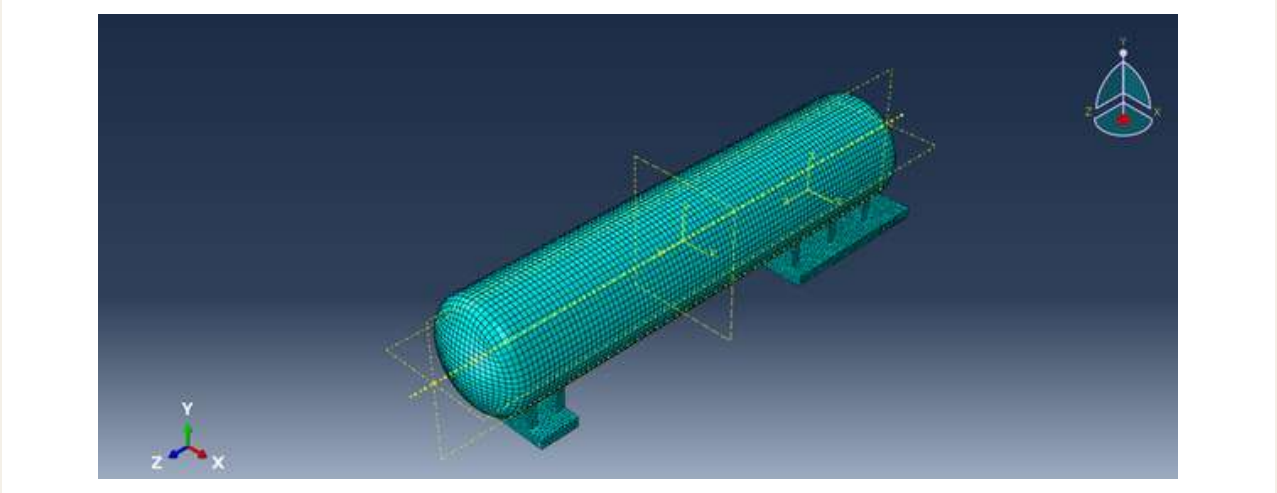
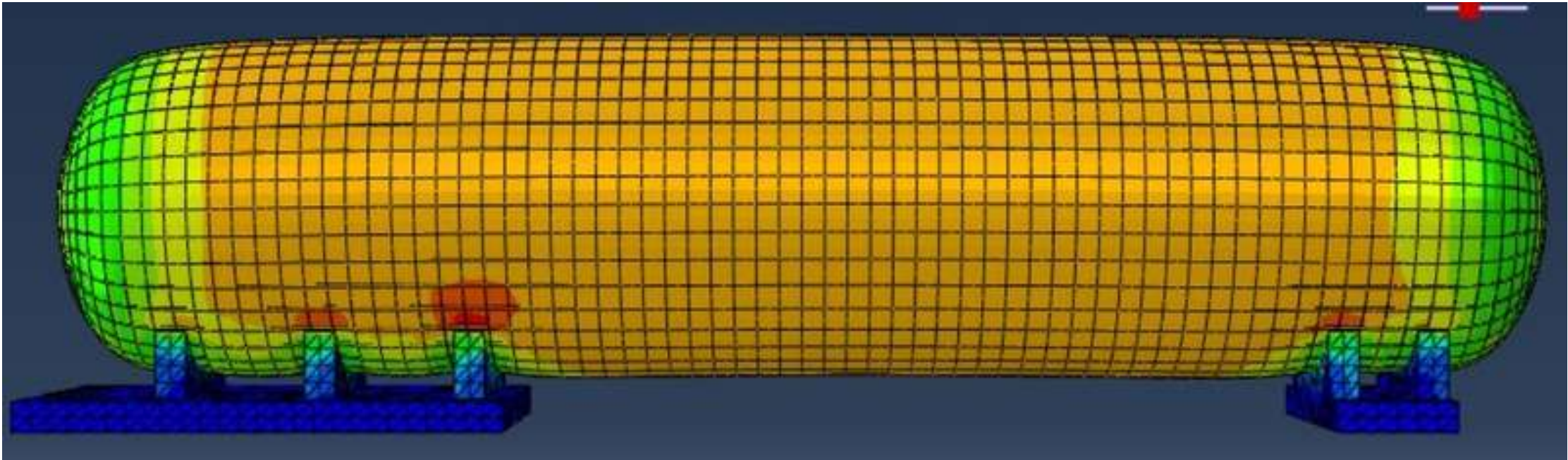
Fuselage internal shear

Envelope limits came first. The critical load factors then drove the distributed-load, shear, and bending calculations used for structural sizing.

Ammonia tanker: strength and crack-growth review

Ammonia tank under internal pressure and road-induced dynamic loading

MY ROLE: SIMULATION AND EXCEL CALCULATIONS



I worked on the simulation and Excel calculations. The model combined 2.2 MPa internal pressure with fixed saddle supports and road-induced dynamic loading. We used a structured mesh for the tank and a finer tetrahedral mesh for the saddle, then carried the critical stress into the crack-growth calculation.

2.2 MPa
internal pressure

308.5 MPa
reported maximum von Mises stress

1.57
reported safety factor

4.95×10^{-5} mm/trip
reported crack-growth rate

262,626 trips
idealized cycles to through-wall leak

The reported 308.5 MPa stress remains below the 485 MPa material strength. However, the safety factor of 1.57 is below the conservative target of 2.0, so the report recommends reviewing the wall thickness.

GOLF BALL IMPACT FEA

Our team modelled a three-layer golf ball in Abaqus using hyperelastic material data from the literature. We compared four club-impact scenarios to see how the contact load moved through the ionomer cover, soft-butadiene mantle, and butadiene core.

4

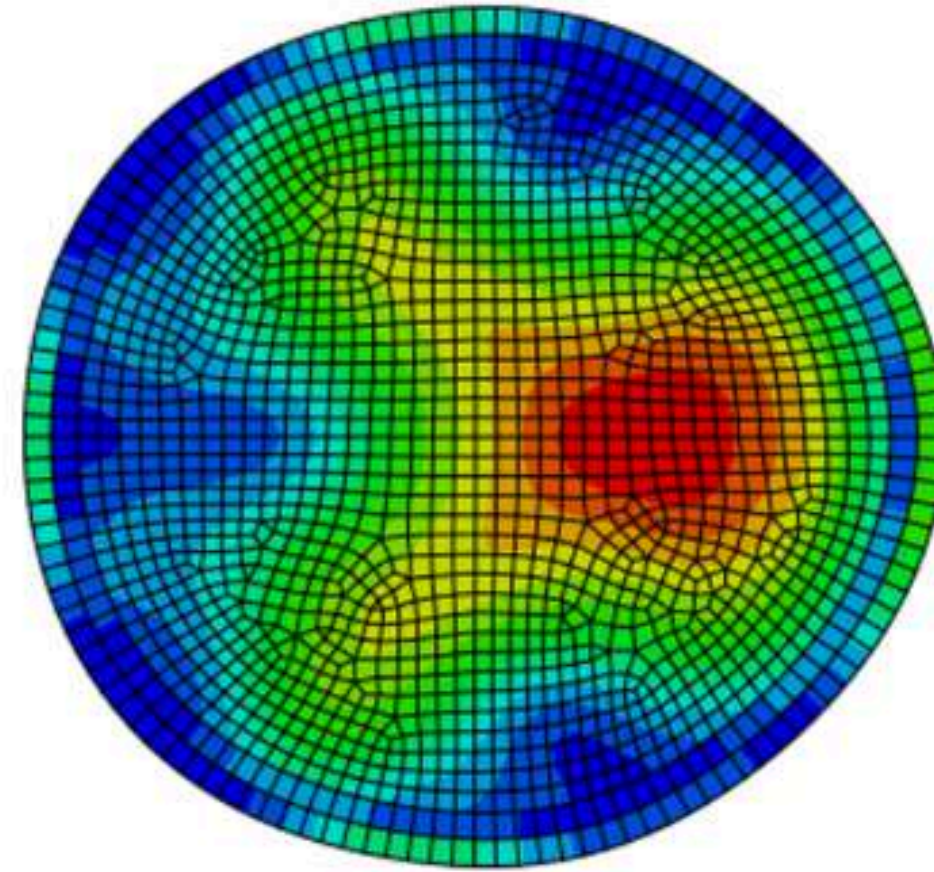
impact scenarios

3

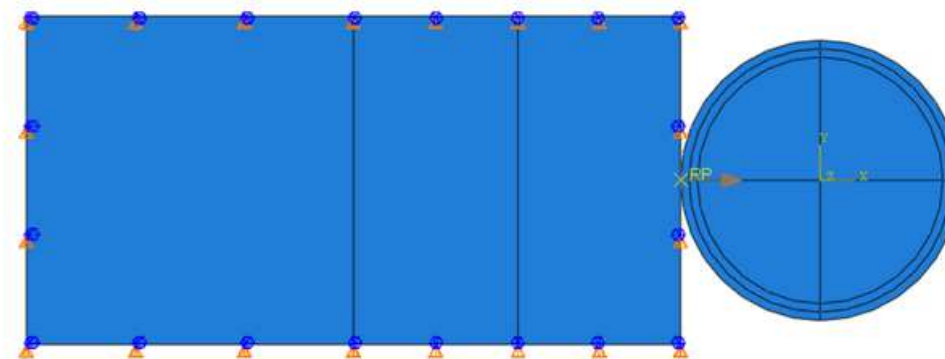
concentric material layers

- Core: butadiene
- Mantle: soft butadiene
- Cover: ionomer

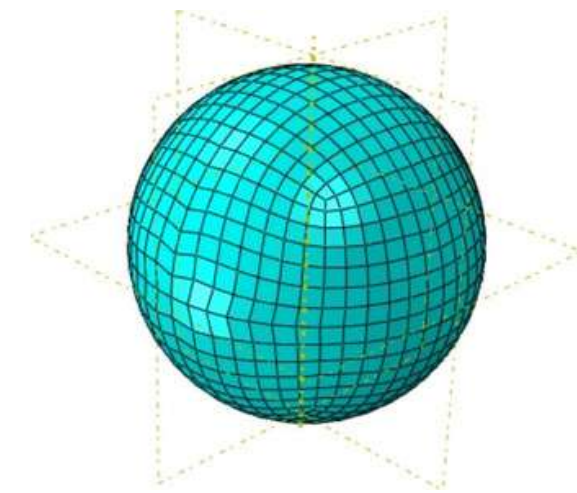
Geometry: 42.67 mm ball diameter, 40.09 mm mantle outer diameter, and 37.53 mm core diameter. Point-mass inertia and a predefined velocity field controlled the impact setup. We reviewed deformation and stress to compare load transfer through the layers.



Stress response

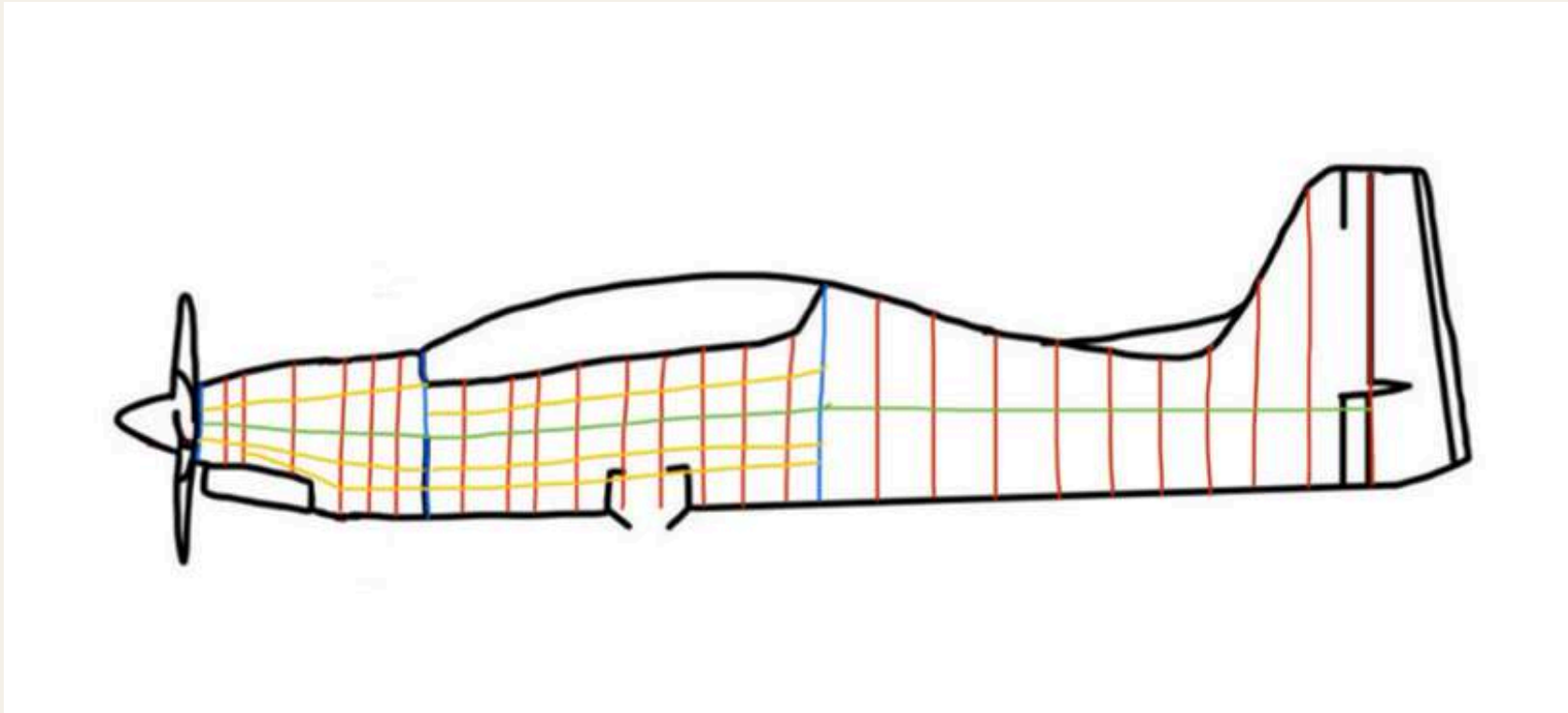


Impact setup



Mesh and layered geometry

JARZ-01 fuselage sizing and buckling iteration



Final fuselage structural layout

MY ASSIGNED CONTRIBUTION: FUSELAGE

I idealized the semi-monocoque fuselage, converted bending and shear loads into panel and stringer stresses, and checked von Mises strength together with local compression and shear buckling. The first sizing did not pass every buckling check, so I iterated the skin and stringer configuration before the final review.

26

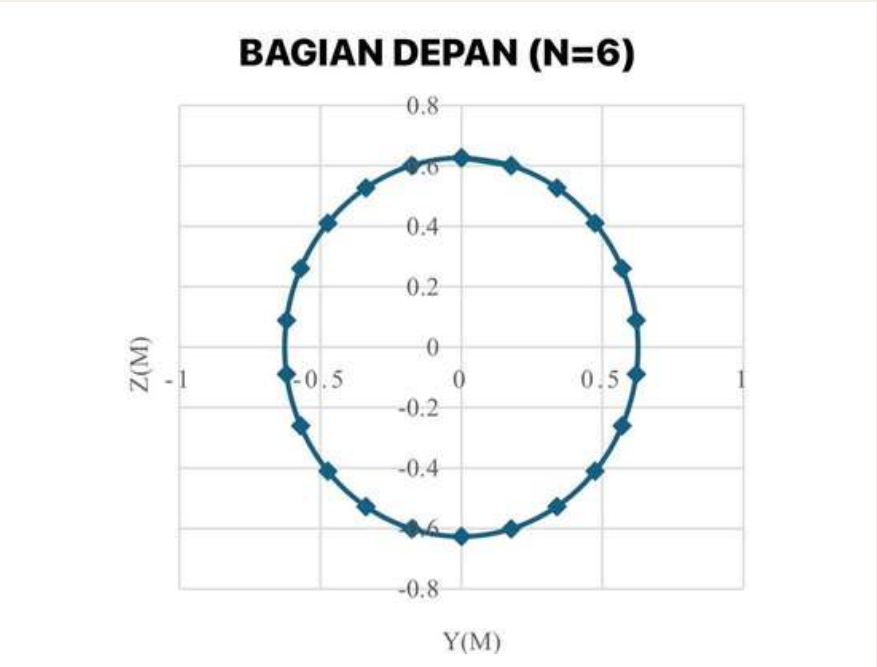
fuselage frames

2

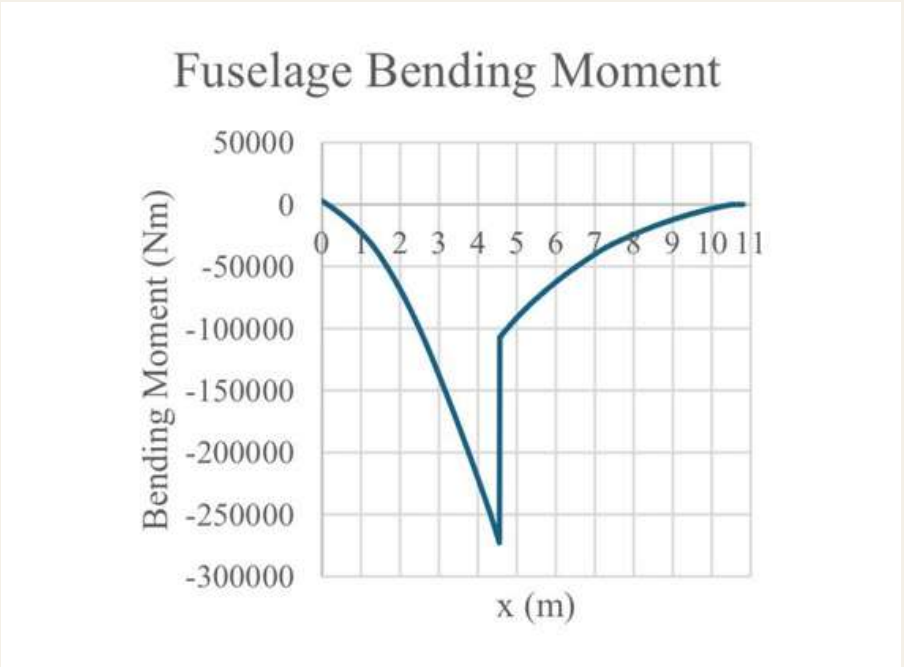
pressure bulkheads

22 / 26 / 24

front / centre / rear stringers



Forward-section idealization



Applied bending distribution

FINAL FUSELAGE CONFIGURATION

The final layout uses Al7075-T6 for the frames and pressure bulkheads, with Al2024-T3 skin and stringers. The report lists different stringer counts for the front, centre, and rear fuselage so reinforcement follows the changing load demand.

TEAM CONTEXT

The complete course project also covered the wing and horizontal tail. The team's wing concept used a 2.6 mm skin with seven stringer pairs and a lug connection to the fuselage.

CABIN CARPET CERTIFICATION PLAN

Our team developed the AeroStox cabin-carpet concept and translated its intended use into a certification basis, compliance evidence, and a verification plan.

CASR 25.853

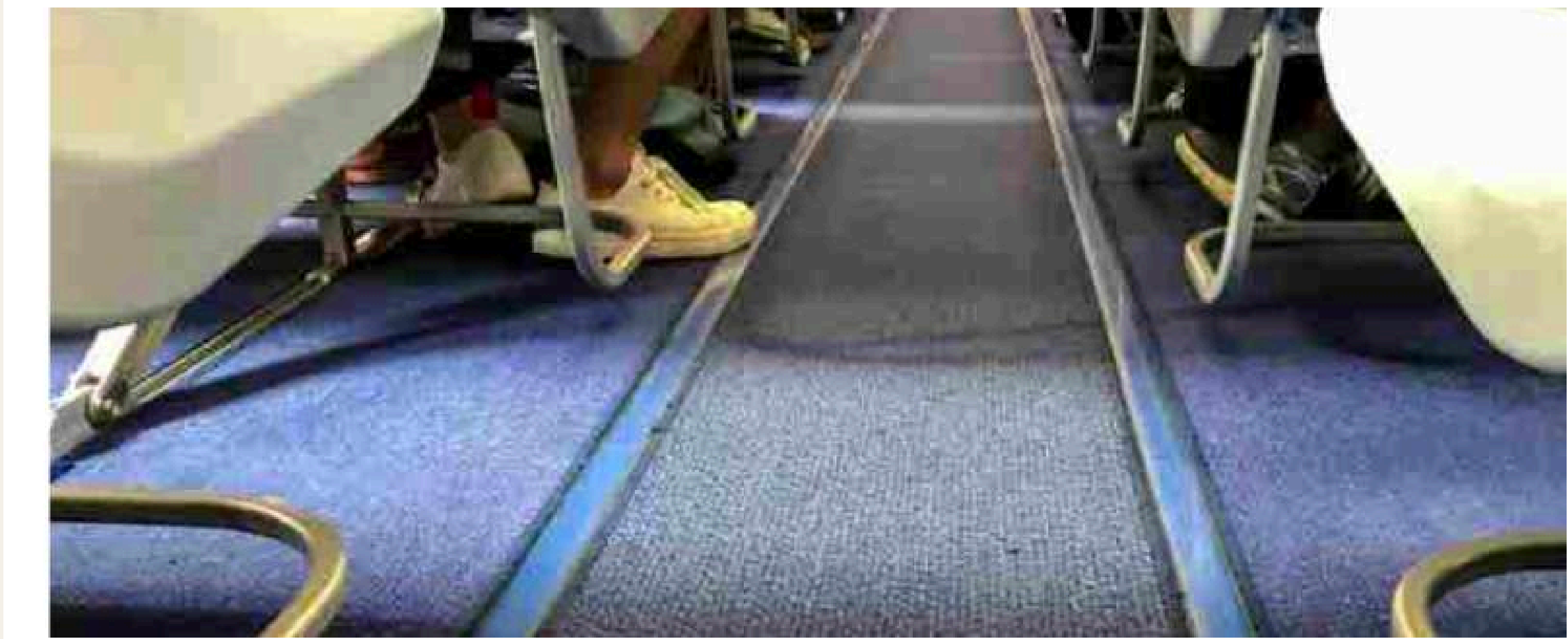
TSO-C72C

ISO 14001

The safety case covers flame propagation, smoke and toxic-gas risk, static electricity, wear, cleaning, and slip resistance. We linked each concern to a planned test or compliance document.

COMPLIANCE OUTPUT

Certification basis, means of compliance, test plan, evidence list, and approval sequence



PLANNED VERIFICATION

- | | | |
|----|---------------------------|---|
| 01 | FIRE AND SMOKE | Vertical Bunsen burner, smoke emission, and toxic-gas tests |
| 02 | SERVICE DURABILITY | Wear, abrasion, stain resistance, and cleaning performance |
| 03 | CABIN FUNCTION | Static electricity, thermal conductivity, and slip resistance |

COMPLIANCE LOGIC

Each hazard is linked to a regulation, an agreed means of compliance, a planned test, and the evidence required for approval. The final output is a certification plan rather than a single pass-or-fail calculation.

Working across teams, not only analysis models

Selected roles in education outreach, student representation, mentoring, and event delivery



Selected documentation from committee and campus activities

SECRETARY GENERAL

Aku Masuk ITB 2024

I coordinated administration, financial and human-resource functions, approval tracking, internal reporting, and institutional compliance for more than 1,200 committee members. The national education-outreach program reached approximately 11,000 high-school students.

HEAD OF EVENTS

Aksi Angkatan Sosial Kemasyarakatan FTMD ITB 2022

I led a three-day social-fundraising campaign with more than 60 participants and approximately IDR 21 million raised.

OTHER RESPONSIBILITIES

Mentor, SONIC KMPN 2024 · Vice External Affairs Representative, FTMD 22 ‘Gemuruh’ · KMPN Senator Team intern · OSKM ITB event staff · Parade Wisuda staff · Olimpiade XI KM ITB staff · HaloTPB! 2022 intern